TOFME: picoDst QA and mixed-event setup for TPC–BTOF matching

Run19 Au+Au $\sqrt{s_{NN}} = 19.6$ GeV, production P24iy (test sample)

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Abstract

This note documents the TOFME project, whose goal is a mixed-event estimate of random (fake) matches between TPC tracks and Barrel Time-of-Flight (BTOF) hits in STAR. The input is the latest Run19 Au+Au 19.6 GeV production (P24iy, library SL24y). First, a QA analysis of the picoDst content was set up and run: event-level, TPC-track-level and TPC–BTOF-association variables, including $1/\beta$ and m^2 versus momentum, local hit coordinates, tray/module occupancy and the TOF matching efficiency. Then the mixed-event machinery was built (Sec. 7): a data-driven BTOF module geometry, projection of the picoDst helix of every track to a BTOF cell, and cell-level matching to hits of the same and of mixed events. For the production-like matching rule (projected cell or its neighbour), a random hit is present in the matching cells for 5.24% of the tracks, equal to the hit occupancy (1M-event sample, 15 runs, 423 775 selected events; stable from run to run once scaled by the occupancy; Sec. 7.4). Finally (Sec. 7.5), β and m^2 are computed for same-event and mixed-event matches in nine centrality classes. The mixed events are overlaid on the track's own event, normalized per projected track, and include hits without a stored time. Random matches are then 4.2% of the TOF-matched tracks in 0–5% and 0.2% in 70–80%, and 2.1% (π), 3.0% (K) and 1.0% (p) of the yield under the PID peaks in 0–5% at $0.8 < p < 1\,\mathrm{GeV}/c$. They explain only a small part of the yield between the proton and deuteron peaks, because the picoDst has no time for BTOF hits not matched to a track. On MuDst (Sec. 7.7), where all hits have a time, the production BTOF matching and calibration re-run on MuDst reproduce production exactly. Two mixed events that use all hits then describe the same-event background between the p and d peaks (SE/ME = 1.29 and 0.92 in 0–5%). Random matches are 2.7% (central) to 6.0% (upper bound) of the TOF matches in 0–5% and 1.6–3.6% in 0–80% (four MuDst files of two runs). **Limitations:** the QA sections (4–6) use a *test sample* of 10000 events from one picoDst file of a single run (20093036); the mixed-event results use 15 runs. No bad-tray list is applied. The time of random hits without a stored time is an assumption in the picoDst sections (Sec. 7.5); the MuDst results use 21160 events only. Statistical uncertainties only. Coloured boxes: **blue** = a choice made and its reason; **red** = caveat / not yet understood; **orange** = decision needed from the analyser.

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1 Overview

Goal of the project. Estimate the probability that a TPC track is matched to a BTOF hit that it did not produce, by pairing tracks with BTOF hits from *different* events of similar topology (mixed events). This first step establishes the data access, software environment and a QA baseline for the quantities the mixed-event study will use.

Main findings (test sample, 5 501 selected events).

- Event selection keeps 5 501 of 10 000 events (55.0%); the $|V_z| < 70\,\text{cm}$ cut dominates the loss (Table 2). 9 989 events carry the minimum-bias trigger 640051.
- Mean refMult is 102.3; mean number of BTOF hits stored per event is 296.5, i.e. 2.94 BTOF hits per TOF-matched primary track (100.8 per event). The number of matched tracks and of BTOF hits are strongly correlated (correlation factor 0.987).
- Track selection (Table 4): 960 157 primary tracks enter the QA sample (174.5 per event); 554 429 of them have a BTOF match with valid β.
- TOF matching efficiency (matched with $\beta > 0$ / TPC tracks) is 0.639 for $0.5 < p_T < 1.5\,\text{GeV}/c$, 0.467 for $0.15 < p_T < 0.3\,\text{GeV}/c$, 0.640 for $|\eta| < 0.5$ and 0.510 for $0.5 < |\eta| < 1$.
- m^2 peaks (for $0.4 < p < 1.0\,\text{GeV}/c$) are at 0.020, 0.244 and 0.877 GeV^2/c^4 for π, K, p, consistent with PDG values within $0.005\,\text{GeV}^2/c^4$.
- 18.7% of matched tracks have `btofMatchFlag`=2 (one hit, several track candidates).
- Mixed events (Sec. 7): the module geometry fitted from production matches reproduces the production projection points to 0.0029 cm. With the picoDst helix, 92.9% of production matches above $1.5\,\text{GeV}/c$ are projected to the same cell. The random-match probability is 5.24% (cell±1, production-like) and 2.34% (cell only), proportional to the number of BTOF hits in the event; it amounts to 6.3% of same-event matches for $0.5 < p_T < 2\,\text{GeV}/c$.
- 1M-event sample (15 runs, Sec. 7.4): $P_{\text{ME}} = 5.24\%$ (cell±1), $f_{\text{rand}} = 6.3\%$ for $0.5 < p_T < 2\,\text{GeV}/c$; per-run P_{ME} divided by the occupancy stays within 0.99–1.01; the geometry refitted on 1M events agrees with the test-run geometry to 0.0005 cm.
- m^2 (Sec. 7.5): with the hit time taken from the picoDst matches and the production path-length formula, the same-event m^2 reproduces production (RMS difference $0.0007\,\text{GeV}^2/c^4$). Mixed events overlaid on the track's own event, normalized per projected track and including hits without stored time, give random matches equal to 4.2% of the TOF matches in 0–5% and 0.2% in 70–80%. In 0–5% at $0.8 < p < 1\,\text{GeV}/c$ they are 2.1% (π), 3.0% (K) and 1.0% (p) of the yield under the peaks. They explain only a small part of the SE yield between the proton and deuteron peaks (SE/ME = 7.41).
- Background origin (Sec. 7.6): the SE excess between the p and d peaks comes from neighbour-cell matches (82% of it), not from dead cells, start time or pile-up. Production picks the largest-ToT hit among the projected and neighbour cells, so random hits can replace the track's own hit. Emulating this choice in picoDst (no ToT) brackets the in-time random matches at 3.3–7.3% of the TOF matches in 0–5%. A late component flat in time, weakly dependent on centrality, has no time in the picoDst and cannot be mixed; this needs the MuDst.
- MuDst (Sec. 7.7, 21160 events of two runs): unmatched BTOF hits are 59% of all hits and much later than matched ones (9.5% vs 1.3% above 26 ns): slow, unreconstructed particles. Re-running `StBTofMatchMaker`/`StBTofCalibMaker` on MuDst reproduces all 16451 production matches of the closure test (time of flight to -14 ps median). Emulation with partner-track claims: random matches 2.7% (0–5%), 1.6% (0–80%), SE/ME between p and d 1.29. Production makers re-run on overlaid hits (partner tracks absent, upper bound): 6.0% and 3.6%, SE/ME 0.92.
- Occupancy: trays 30, 34, 102 have no hits at all; 140 of 3840 modules (3.6%) are empty. No matched track points to an empty module.

Choices made (one line each; details where used).

1. Production P24iy (latest of P21ic, P23id, P24iy in the file catalog) — Sec. 2.
2. Input read through XRootD from the `local` storage replicas; the NFS catalog replicas checked were absent on disk — Sec. 2.
3. STAR library SL24y run inside the SL7 container on the AlmaLinux 9 interactive node — Sec. 2.2.
4. Minimum-bias trigger IDs 640001/11/21/31/41/51 — Sec. 3.1.
5. Event cuts $|V_z| < 70\,\text{cm}$, $V_r < 2\,\text{cm}$; no $V_z^{\text{TPC}} - V_z^{\text{VPD}}$ cut — Sec. 3.1.
6. Track cuts: primary, $p_T > 0.15\,\text{GeV}/c$, nHitsFit≥ 15, nHitsFit/nHitsMax> 0.52, gDCA$< 3\,\text{cm}$, $|\eta| < 1$ — Sec. 3.2.
7. TOF match: `btofMatchFlag`> 0 and $\beta > 0$; primary momentum used for m^2— Sec. 3.3.
8. Approximate TOF $n\sigma$ with a fixed $1/\beta$ resolution of 0.013 (QA only) — Sec. 3.3.
9. BTOF module geometry fitted from production matches (no `StBTofGeometry`) — Sec. 7.
10. Projection with the picoDst global helix; matching at cell level, "cell" and "cell$\pm$1" — Sec. 7.
11. Mixing within the same run, 14 V_z $\times$ 9 refMult classes, 5 partner events — Sec. 7.
12. Large sample: one job per run ($\geq$60k events), bad runs excluded, catalog replicas deduplicated — Sec. 7.4.
13. Centrality from StRefMultCorr (2019 19.6 GeV), pile-up events excluded from m^2 — Sec. 7.5.
14. Random m^2 from overlay mixed events (partner hits added to the track's own event), normalized per projected track, all hits included (untimed hits assumed to have the timed-hit time distribution) — Sec. 7.5.

2 Dataset, software and computing

2.1 Dataset

The STAR file catalog was queried with `get_file_list.pl` for `filetype=daq_reco_picoDst, trgsetupname=production_19GeV_2019`. Three productions exist: P21ic (SL21c), P23id (SL23d) and P24iy (SL24y).

Choice

Use P24iy, the most recent production of the Run19 19.6 GeV data (library SL24y). The picoDst format is the one defined in `StRoot/StPicoEvent` of https://github.com/star-bnl/star-sw; the SL24y copy of that package is used to read it.

For P24iy the catalog lists 100 560 files on HPSS, 99 273 on `local` storage (1.42×10^9 events) and 100 958 NFS entries. For the latest runs, the NFS paths checked (first 3000 entries ordered by run number) do not exist on disk.

Choice

Read the `local` replicas through XRootD, `root://xrdstar.rcf.bnl.gov:1095//home/starlib/...` (absolute path, double slash). The full list is `filelists/P24iy_19GeV_all_xrd.list`; the catalog dump is `filelists/local_P24iy_all.txt`.

Test sample. One file from the last run in the catalog, run 20093036: `st_physics_20093036_raw_1000008.picoDst.root` (17041 events in the TTree). The first 10000 events were processed (`filelists/test.list`).

Caveat

A single file of a single run is not representative of the whole dataset (beam conditions, detector status, luminosity vary run by run). Statistics are adequate for shapes and mean values, not for run-dependent or tray-dependent conclusions.

2.2 Software and computing

Table 1: Software environment, from `logs/provenance.txt` (version banners on the node) and the run log.

Node	starsub01.sdcc.bnl.gov, AlmaLinux release 9.8 (Olive Jaguar)
Container	`/cvmfs/star.sdcc.bnl.gov/containers/rhic_sl7_05_26.sif` (Scientific Linux release 7.3 (Nitrogen))
STAR library	SL24y (`/star/nfs4/AFS/star/packages/SL24y`)
ROOT	5.34/38 (STAR spack build)
Compiler	gcc 4.8.5
Executable	`root4star` (SL24y, .sl73_x8664_gcc485)
Run date	Tue Nov 3 22:14:23 2026 (EST)

Choice

The interactive nodes run AlmaLinux 9 while the STAR libraries are SL7 builds. All builds (`cons`) and jobs run in the SL7 container via the wrapper `scripts/sl7.sh` (`apptainer exec -B /gpfs01,/gpfs,/star,/cvmfs,/direct,/sdcc ... tcsh -c "starver SL24y; ..."`).

Working directory: `/gpfs01/star/pwg/dongx/tof/TOFME`. Code:

- `StRoot/StTofMEQAMaker/StTofMEQAMaker.{h,cxx}` — QA maker (`StMaker`) behind `StPicoDstMaker` in read mode.
- `macros/runTofMEQA.C` — chain macro; sets cuts and trigger IDs.
- `scripts/run_qa.shNlistoutlog` — runs the macro in the container.
- `macros/drawTofMEQA.C`, `scripts/draw_qa.sh` — quick-look plots on the node.
- `StRoot/StTofMEMaker`, `macros/runTofME.C`, `macros/makeGeom.C`, `scripts/run_me.sh`, `scripts/me_test.sh`, `scripts/select_runs.py`, `scripts/me_v3.sub`, `scripts/merge_v3.sh` — projection, mixed-event and m^2 step (Sec. 7–7.5).
- Note figures/tables: `analysis_note/scripts/make_figures.C`, `make_numbers.C`, `make_me.C`, `make_me_runs.C`, `make_m2.C`, `parse_log.py`, run by `build.sh` on the local copy `analysis_note/data/qa_test_10k.root`.

Choice

Only the picoDst branches `Event`, `Track`, `BTofHit` and `BTofPidTraits` are enabled (`StPicoDstMaker::SetStatus`), since nothing else is used.

Performance. Processing 10000 events took 324 s wall time but only 13.7 s CPU; 99.4% of the wall time is spent in `StPicoDstMaker`, i.e. the job is limited by XRootD I/O, not by the analysis.

Caveat

At this rate (about 30 events/s per job) full-dataset processing needs batch jobs, ideally scheduled close to the data; this has not been set up yet.

3 Method and definitions

The QA maker loops over picoDst events; per event it (i) records trigger IDs and the vertex before any cut, (ii) applies the event selection, (iii) fills event-level and BTOF-hit histograms, (iv) loops over tracks applying the track selection and fills track and TOF histograms. All histograms are written to one ROOT file (`output/qa_test_10k.root`).

3.1 Event selection

Cuts are applied in the order listed; the cut flow is stored in `hEvtCut`.

1. **Trigger:** the event fires at least one of 640001, 640011, 640021, 640031, 640041, 640051 (`StPicoEvent::isTrigger`).
2. **Vertex z:** $|V_z| < 70\,\mathrm{cm}$, with V_z the TPC primary vertex (`primaryVertex()`).
3. **Vertex radius:** $V_r = \sqrt{V_x^2 + V_y^2} < 2\,\mathrm{cm}$, measured from the origin of the STAR frame (no beam-spot offset).
4. $|V_z - V_z^{\mathrm{VPD}}|$: *not applied* (cut value set negative), only histogrammed.

Choice

The trigger list is taken as the Run19 19.6 GeV minimum-bias offline trigger IDs (640001–640051 in steps of 10). This list was *not* checked against the trigger database; only 640051 occurs in the test file (Table 3), so the other IDs are untested here.

Choice

$|V_z| < 70\,\mathrm{cm}$ keeps the vertex well inside the TPC and BTOF acceptance and is a typical value for collider-mode BES-II analyses. $V_r < 2\,\mathrm{cm}$ rejects beam-pipe interactions. Both are QA defaults, not optimised.

Caveat

The $V_z^{\mathrm{TPC}} - V_z^{\mathrm{VPD}}$ cut, often used against pile-up, is not applied because at 19.6 GeV the VPD efficiency and resolution are limited (Sec. 4). Pile-up rejection (e.g. refMult versus nBTOFMatch) is also not applied yet.

3.2 Track selection

Tracks are taken from the picoDst track array; for each track:

1. **Primary** track (`isPrimary()`, i.e. $|\vec{p}_{\mathrm{prim}}| > 0$), $p_T^{\mathrm{prim}} > 0.15\,\mathrm{GeV}/c$. Histograms of nHitsFit, nHitsFit/nHitsMax and gDCA are filled here, *before* the next cuts.

2. **Quality:** nHitsFit ≥ 15, nHitsFit/nHitsMax > 0.52 (rejects split tracks), gDCA $< 3\,\mathrm{cm}$ with gDCA $= |\vec{\mathrm{origin}}_{\mathrm{global}} - \vec{V}|$ (`gDCA(pv).Mag()`). The η histogram is filled here.

3. **Acceptance:** $|\eta| < 1$ (BTOF acceptance). All remaining track and TOF histograms use this sample, which is also the denominator of the matching efficiency.

Track kinematics (p, p_T, η, ϕ) use the *primary* momentum, and the charge q is the sign of nHitsFit as stored in the picoDst.

3.3 TPC–BTOF association variables

A track is *associated* with BTOF if it has a `StPicoBTofPidTraits` entry (`isTofTrack()`). From it the following are used:

- `btofMatchFlag`: 0 no match, 1 one-to-one, 2 one hit with several track candidates. **Matched** means flag > 0.
- `btofCellId` $= (\mathrm{tray}-1)\cdot 192 + (\mathrm{module}-1)\cdot 6 + (\mathrm{cell}-1)$; tray $\in [1,120]$, module $\in [1,32]$, cell $\in [1,6]$.
- y_{local}, z_{local}: local coordinates of the track projection relative to the centre of the matched cell; y is along the row of six pads in a module (nominal pad pitch 3.15 cm), z is along the beam (nominal pad length 6.1 cm). The production matching (`StBTofMatchMaker`, SL24y) requires $|z_{\mathrm{local}}| < 3.05\,\mathrm{cm}$ (`mZLocalCut`, checked in the source).
- global hit position `btofHitPos()` (used for the ϕ–z map);
- $\beta =$ `btofBeta()`. Matched tracks without valid timing have $\beta = 0$.

Derived quantities, with p the primary momentum:

$$m^2 = p^2\left(\frac{1}{\beta^2}-1\right), \qquad \frac{1}{\beta^i_{\mathrm{exp}}} = \frac{\sqrt{p^2+m_i^2}}{p}, \qquad n\sigma_i^{\mathrm{TOF}} = \frac{1/\beta - 1/\beta^i_{\mathrm{exp}}}{0.013}, \quad i = \pi, \mathrm{K}, \mathrm{p}. \tag{1}$$

The **TOF-matched sample** (numerator of the efficiency) requires flag > 0 and $\beta > 0$. The matching efficiency is $\varepsilon_{\mathrm{match}} = N_{\mathrm{TOF}}/N_{\mathrm{TPC}}$, with binomial (Bayesian, uniform prior, 68% interval) errors in the figures and $\sqrt{\varepsilon(1-\varepsilon)/N}$ in the tables.

Caveat

$n\sigma^{\mathrm{TOF}}$ here uses a constant $1/\beta$ resolution of 0.013 for orientation only; it is not the calibrated $n\sigma$ stored in `StPicoBTofPidTraits`. $\varepsilon_{\mathrm{match}}$ is the product of the BTOF geometric acceptance, the hit efficiency and the matching efficiency, and still contains random matches — separating those is the aim of the mixed-event step.

BTOF hits. For every stored `StPicoBTofHit` the tray, module and cell are decoded from `id()` with the encoding above. Note that `StPicoBTofHit::cell()` in SL24y returns `mId/6+1` rather than the cell number; the maker computes `id\%6+1` itself. The picoDst stores only the cell ID of each hit, not its position or time.

4 Event-level QA

All figures in Secs. 4–6 are produced by `analysis_note/scripts/make_figures.C` from `data/qa_test_10k.root` (copy of `output/qa_test_10k.root` on the node). Unless stated otherwise they show the 5 501 selected events; the event count printed above each frame is the number of selected events. Errors are statistical.

Table 2: Event cut flow for the test sample (`hEvtCut`; generated table `tables/evtcut.tex`).

Cut (cumulative)	Events	Fraction of previous	Fraction of all
All events read	10 000	1.000	1.000
Minimum-bias trigger	9 989	0.999	0.999
$\lvert V_z\rvert < 70$ cm	5 786	0.579	0.579
$V_r < 2$ cm	5 501	0.951	0.550
$\lvert V_z - V_z^{\mathrm{VPD}}\rvert$ (not applied)	5 501	1.000	0.550

Table 3: Trigger IDs stored in the picoDst for all 10 000 events read (an event can carry several). IDs with fewer than 50 events are summed. 21 distinct IDs occur in total (full list in `logs/numbers.txt`).

Trigger ID	Events	Fraction
640051	9 989	0.9989
6	9 371	0.9371
36	8 785	0.8785
35	8 437	0.8437
37	3 669	0.3669
640601	303	0.0303
640039	20	0.0020
Other (14 IDs, each < 50)	59	0.0059

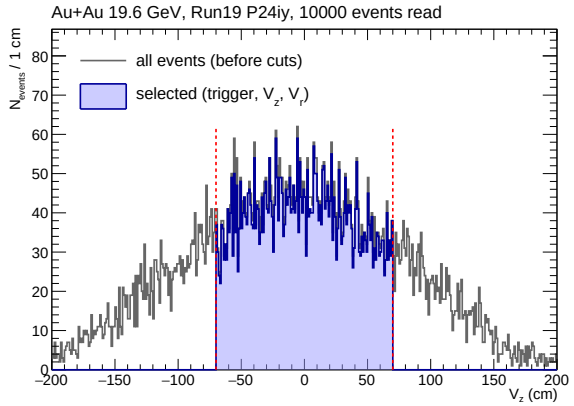
Trigger. 9 989 of 10 000 events carry 640051, the only minimum-bias ID present. IDs 6, 35, 36 and 37 also appear in a large fraction of events.

Caveat

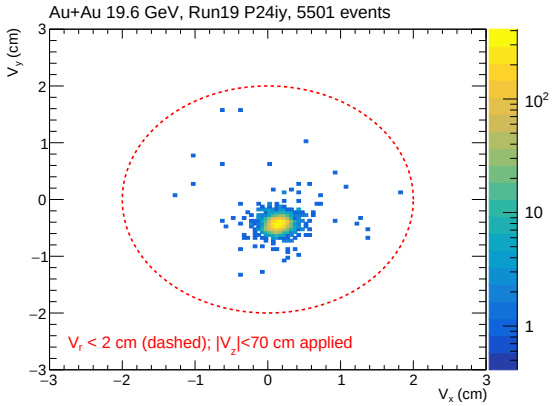
The meaning of trigger IDs 6, 35, 36, 37, 640601 and 6408xx was not looked up. They do not affect the selection, which only requires one of the minimum-bias IDs.

Vertex. The V_z distribution before cuts is broad (RMS 80.9 cm, mean -8.0 cm), so the $\lvert V_z\rvert < 70$ cm cut is the largest loss (Table 2, Fig. 1a). The transverse vertex is at $(\langle V_x\rangle, \langle V_y\rangle) = (0.149, -0.421)$ cm with RMS 0.105 and 0.100 cm (Fig. 1b), well inside the $V_r < 2$ cm circle centred at the origin.

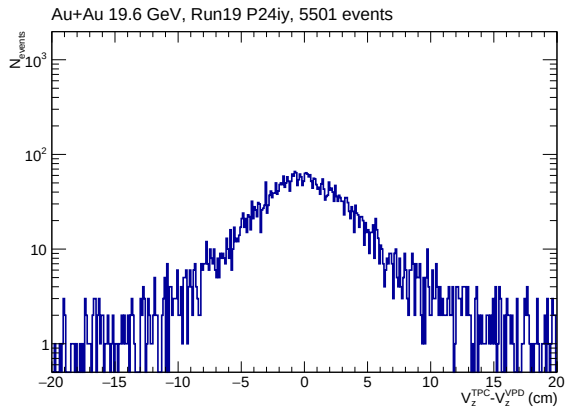
VPD vertex. For 92.9% of the selected events V_z^{VPD} lies within ± 200 cm; the remainder has no valid VPD vertex. These events, together with events with $\lvert\Delta V_z\rvert > 20$ cm, populate the under-/overflow of Fig. 1c (117 / 530 events). A Gaussian fit to the core ($\lvert\Delta V_z\rvert < 3$ cm) of $\Delta V_z = V_z - V_z^{\mathrm{VPD}}$ gives mean -0.14 cm and $\sigma = 2.71$ cm; only 51.2% of the selected events have $\lvert\Delta V_z\rvert < 3$ cm.



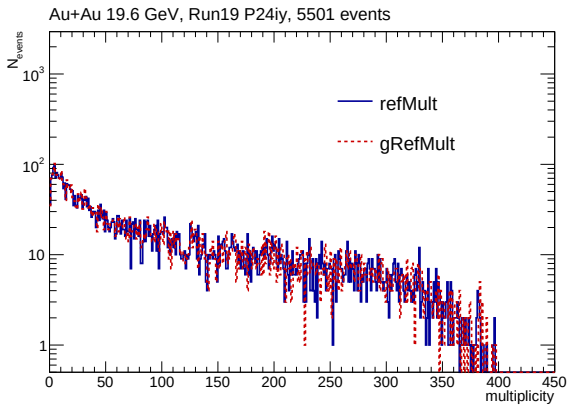
(a) V_z for all 10 000 events read (grey line) and for the selected events (blue, filled). Red dashed lines: $|V_z| < 70$ cm. 1 cm bins.



(b) V_y vs V_x for selected events; the red dashed circle is $V_r = 2$ cm. 0.05 cm bins.



(c) $V_z^{\mathrm{TPC}} - V_z^{\mathrm{VPD}}$ (0.1 cm bins, log scale). The central Gaussian core has $\sigma = 2.71$ cm; events without VPD vertex are outside the range.



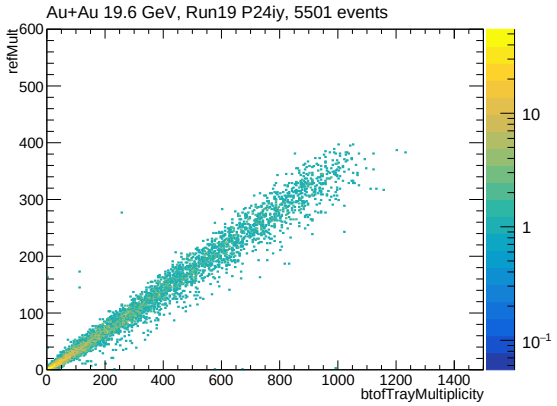
(d) refMult (blue solid) and gRefMult (red dashed), 1-unit bins, log scale. Means 102.3 and 102.4.

Figure 1: Event-level distributions of the test sample.

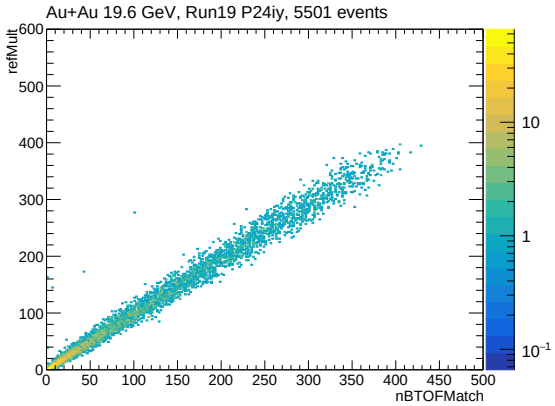
Multiplicities and BTOF hits. Per selected event the mean refMult is 102.3, the mean nBTOFMatch (`StPicoEvent::nBTOFMatch`) is 104.1 and the mean number of stored BTOF hits is 296.5 (maximum 1230). The mean of `btofTrayMultiplicity` is also 296.5. Figure 2c shows that the number of BTOF hits grows linearly with the number of matched tracks, and Fig. 2d that the number of TOF-matched *primary* QA tracks follows it closely (correlation factor 0.987). On average there are 2.94 BTOF hits per TOF-matched QA track; most hits are therefore *not* associated with a selected track — they come from tracks outside the selection, secondaries, or noise, and are the candidates for random matches.

Caveat

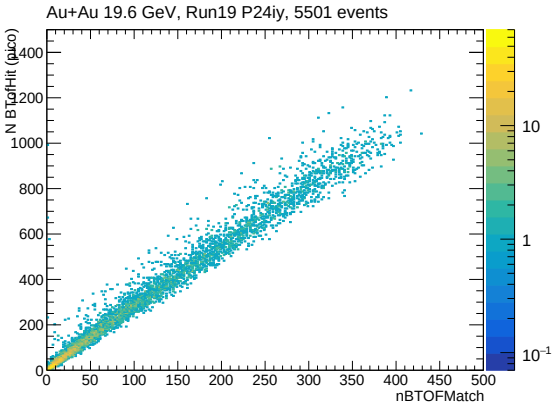
The hit counts and `btofTrayMultiplicity` have identical mean and RMS in this sample, which suggests the picoDst keeps all BTOF hits of the event. Not verified in `StPicoDstMaker`.



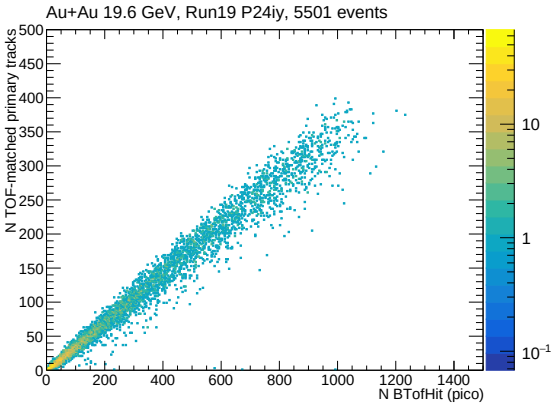
(a) refMult vs `btofTrayMultiplicity`.



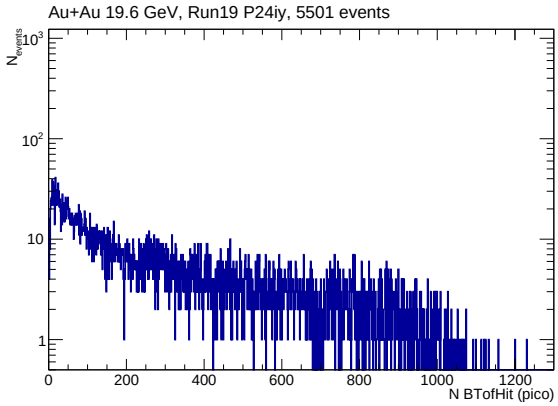
(b) refMult vs nBTOFMatch. No pile-up cut is applied; the band is narrow with few outliers in this sample.



(c) Number of stored BTOF hits vs nBTOFMatch.



(d) Number of TOF-matched primary QA tracks per event (Sec. 3.3) vs number of stored BTOF hits.



(e) Number of stored BTOF hits per selected event, log scale.

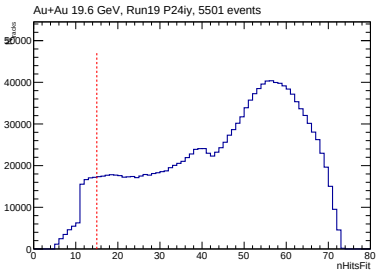
Figure 2: Event multiplicity correlations (log colour scale for 2D panels).

5 Track-level QA

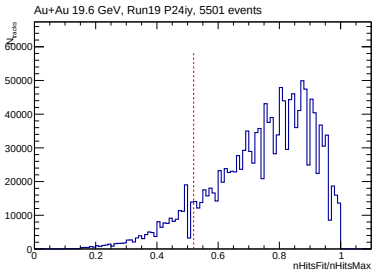
Table 4: Track cut flow, summed over the 5 501 selected events (generated table `tables/trkcut.tex`). Rows 1–3 are the TPC selection of Sec. 3.2; rows 4–6 the BTOF association of Sec. 3.3.

Selection (cumulative)	Tracks	Per event	Fraction of previous
Primary tracks, $p_T > 0.15\,\mathrm{GeV}/c$	1 542 971	280.5	–
+ nHitsFit≥ 15, nHitsFit/nHitsMax> 0.52, gDCA$< 3\,\mathrm{cm}$	1 347 381	244.9	0.873
+ $\lvert\eta\rvert < 1$ (QA track sample, denominator)	960 157	174.5	0.713
+ has BTofPidTraits	562 313	102.2	0.586
+ btofMatchFlag> 0	562 313	102.2	1.000
+ $\beta > 0$ (TOF-matched sample, numerator)	554 429	100.8	0.986

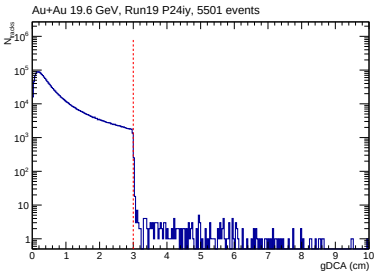
Table 4 gives the track yields. The quality cuts keep most primary tracks, and the $|\eta| < 1$ cut keeps 71.3% of those: with the inner TPC (iTPC) upgrade in Run19, primary tracks extend well beyond $|\eta| = 1$ (Fig. 3d). The QA sample has 174.5 tracks per event, mean $p_T = 0.516\,\mathrm{GeV}/c$.



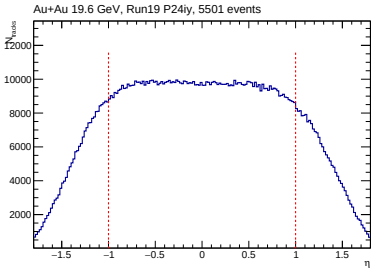
(a) nHitsFit, primary $p_T > 0.15\,\mathrm{GeV}/c$, before quality cuts. Red: cut at 15.



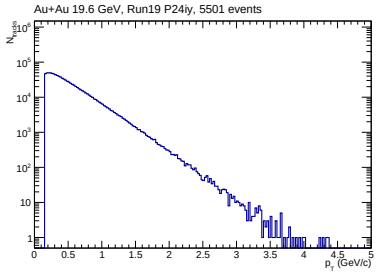
(b) nHitsFit/nHitsMax, same sample. Red: cut at 0.52. The spikes are a binning effect of the ratio of two integers.



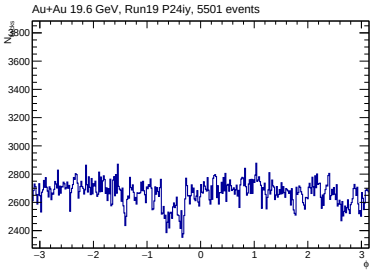
(c) gDCA, same sample, log scale. Red: cut at 3 cm.



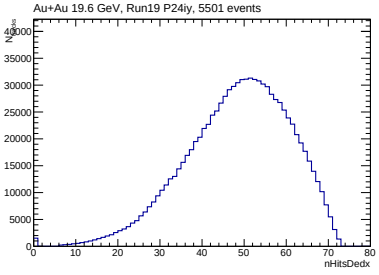
(d) η after quality cuts. Red: $|\eta| < 1$.



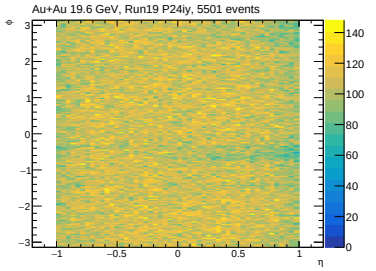
(e) p_T of QA tracks, log scale.



(f) ϕ of QA tracks (1° bins). Flat within about ±5% except a dip near $\phi \approx -0.6$ to -0.3.

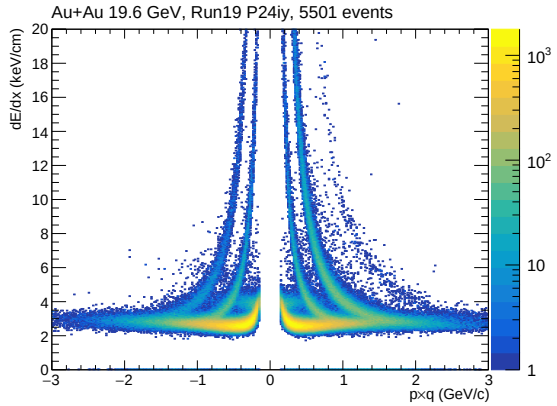


(g) nHitsDedx of QA tracks.

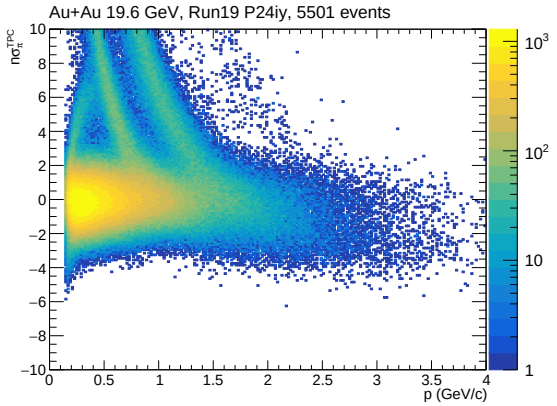


(h) ϕ vs η of QA tracks (linear colour scale). A region of lower density is visible at $\phi \approx -0.6$ for $\eta > 0$.

Figure 3: TPC track QA. Top row: distributions of the cut variables (before the cuts) and η; other panels: QA track sample (primary, $p_T > 0.15\,\mathrm{GeV}/c$, quality cuts, $|\eta| < 1$).



(a) dE/dx (keV/cm, as stored in the picoDst) vs $p \times q$ for QA tracks, log colour scale. π, K and p bands are separated at low momentum for both charges; a deuteron band is visible for positive charge.



(b) TPC $n\sigma_\pi$ vs p of QA tracks, log colour scale.

Figure 4: TPC dE/dx PID for the QA track sample.

Caveat

The reduced track density near $\phi \approx -0.6$, $\eta > 0$ (one TPC sector region of the west half, by its position) was seen by eye only and not traced to a sector or run condition. The ϕ and η–ϕ structure was not quantified.

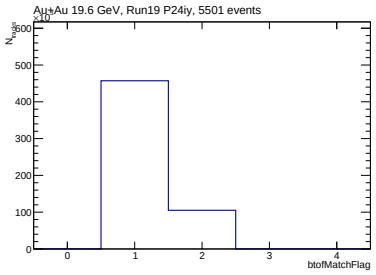
6 TPC–BTOF association QA

Match flag and β. Every QA track with a `BTofPidTraits` entry has `btofMatchFlag`> 0 (0 entries with flag 0), so the picoDst keeps only matched tracks. 457 193 (81.3%) are one-to-one matches and 105 120 (18.7%) have flag 2, i.e. the hit had more than one track candidate (Fig. 5a). 7 884 matched tracks (1.4%) have $\beta = 0$ (no valid time) and are dropped from the TOF-matched sample. The β distribution (Fig. 5b) peaks just below 1 with a tail above 1 containing 3.72% of the matched tracks; these give $m^2 < 0$.

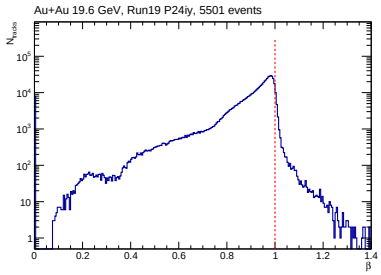
Local coordinates. z_{local} is flat with sharp edges at ± 3.05 cm, the matching window of `StBTofMatchMaker` (Fig. 5e). y_{local} (Fig. 5d) has a plateau over the pad, $|y_{\mathrm{local}}| \lesssim 1.6$ cm, falling over about 1 cm on each side, and flat tails out to ± 5 cm about a factor 50 below the plateau (read by eye from the figure); 92.4% of the entries are within $|y_{\mathrm{local}}| < 1.8$ cm, RMS 1.21 cm. The tails broaden at low momentum, where multiple scattering and the projection uncertainty are largest (Fig. 5f).

Caveat

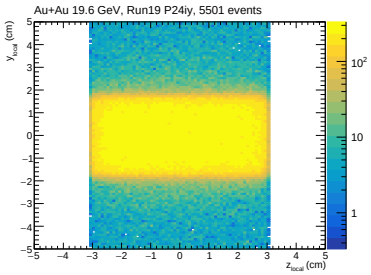
The flat y_{local} tails are the region where random matches are expected to dominate. Their size relative to the plateau is a direct handle for the mixed-event normalisation.



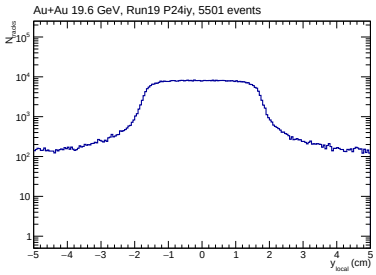
(a) `btofMatchFlag` of QA tracks with BTofPidTraits.



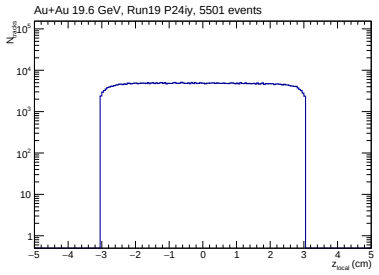
(b) β of matched tracks, log scale; red dashed line: $\beta = 1$. The $\beta = 0$ entries are in the first bin.



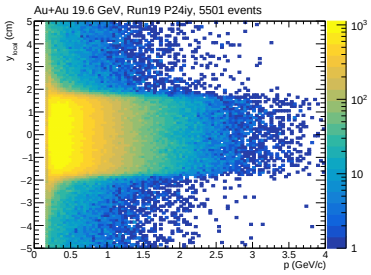
(c) y_{local} vs z_{local}, log colour scale.



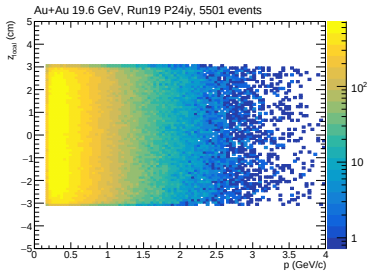
(d) y_{local} of matched tracks, 0.05 cm bins, log scale.



(e) z_{local} of matched tracks, 0.05 cm bins, log scale.



(f) y_{local} vs p, log colour scale.



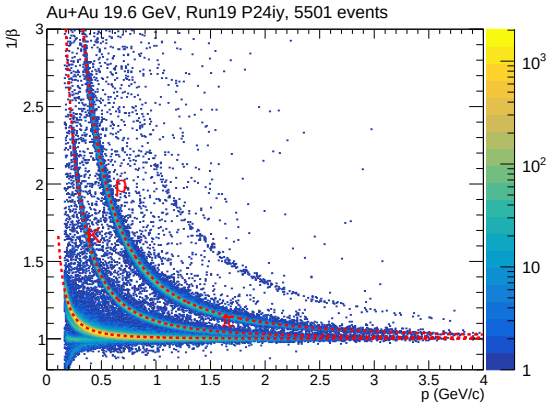
(g) z_{local} vs p, log colour scale.

Figure 5: Match flag, β and local hit coordinates for QA tracks matched to BTOF (flag>0).

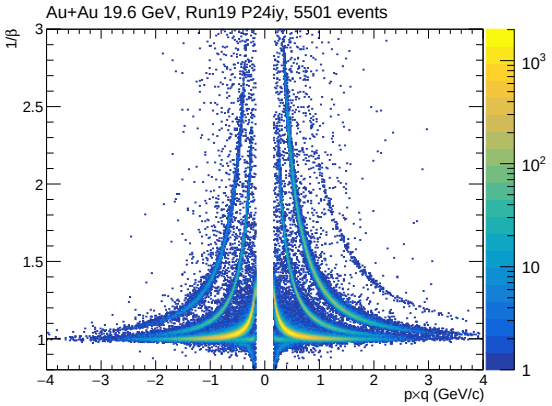
Particle identification. $1/\beta$ vs p (Fig. 6a) shows clean π, K and p bands that follow the expected curves, plus a deuteron band and a uniform background between bands. Fitting Gaussians to the m^2 projection for $0.4 < p < 1.0\,\mathrm{GeV}/c$ (Fig. 6e) gives

	mean (GeV^2/c^4)	σ (GeV^2/c^4)	PDG m^2
π	0.020	0.008	0.019
K	0.244	0.012	0.244
p	0.877	0.034	0.880
d	3.332	0.407	3.520

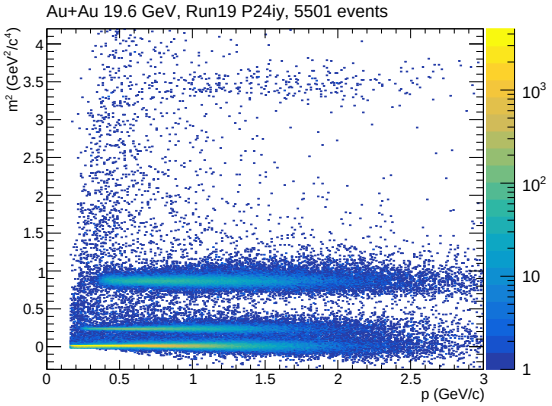
The π, K and p positions agree with PDG values within $0.005\,\mathrm{GeV}^2/c^4$, i.e. the TOF calibration in P24iy is in good shape for this run. The deuteron fit has low statistics in this momentum window and a broad, asymmetric peak; it is shown for orientation only.



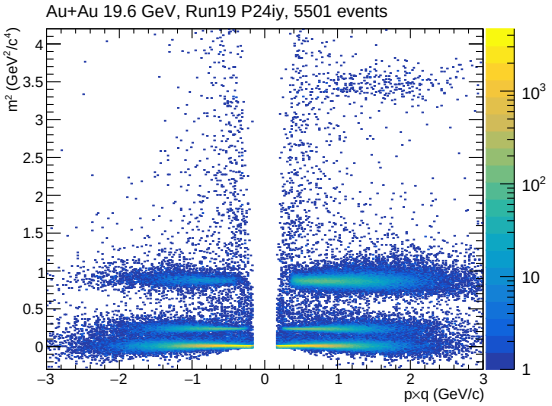
(a) $1/\beta$ vs p; red dashed curves are the expectations for π, K, p.



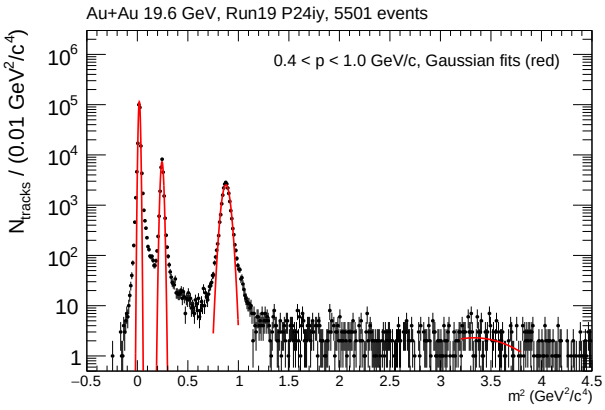
(b) $1/\beta$ vs $p \times q$.



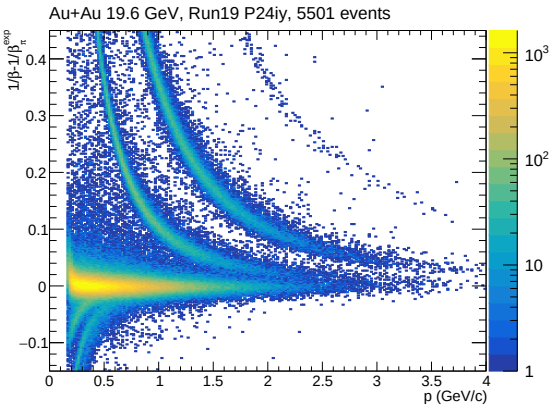
(c) m^2 vs p.



(d) m^2 vs $p \times q$; d appears only for positive charge.

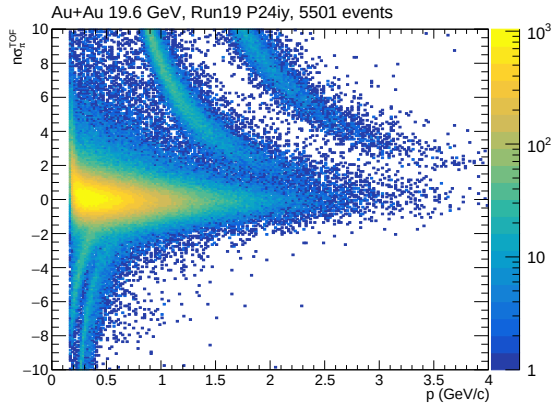


(e) m^2 for $0.4 < p < 1.0\,\mathrm{GeV}/c$ (black points), with Gaussian fits around the π, K, p and d peaks (red). Log scale; $0.01\,\mathrm{GeV}^2/c^4$ bins.

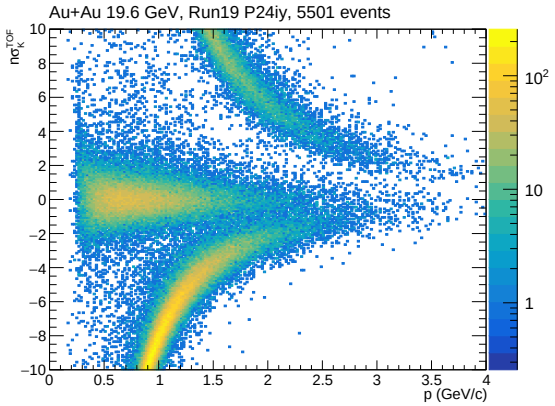


(f) $1/\beta - 1/\beta^{\pi}_{\mathrm{exp}}$ vs p.

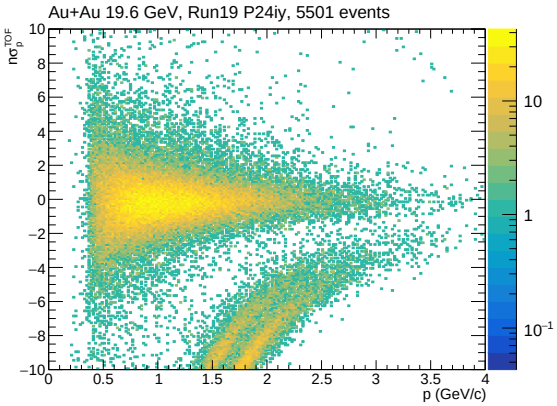
Figure 6: TOF PID for the TOF-matched sample (flag> 0, $\beta > 0$), log colour scale.



(a) $n\sigma_{\pi}^{\mathrm{TOF}}$ vs p.



(b) $n\sigma_{\mathrm{K}}^{\mathrm{TOF}}$ vs p.



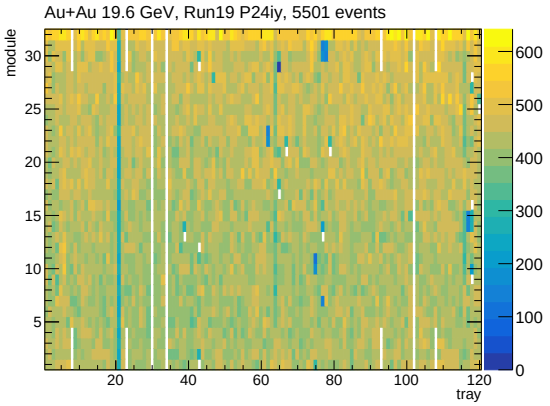
(c) $n\sigma_{\mathrm{p}}^{\mathrm{TOF}}$ vs p.

Figure 7: Approximate TOF $n\sigma$ with a fixed $1/\beta$ resolution of 0.013 (Sec. 3.3); not calibrated.

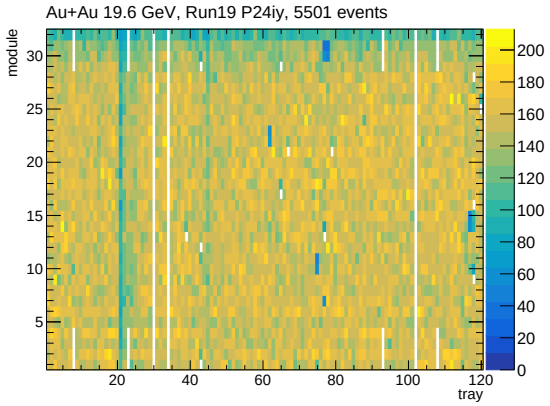
Occupancy. Figure 8a shows the tray–module occupancy of all stored hits. The median module has 440 hits in this sample. Trays 30, 34, 102 are empty; trays with empty modules, given as tray(number of empty modules), are 8 (8), 23 (8), 39 (1), 43 (3), 65 (1), 67 (1), 77 (1), 79 (1), 93 (8), 108 (8), 118 (3), 120 (1). In total 140 of 3840 modules (3.6%) are empty, and no matched track points to an empty module (0 entries), as expected. In each of the four trays with 8 empty modules, the empty modules are 1, 2, 3, 4, 29, 30, 31, 32 (same list in all four). Isolated low-occupancy modules are also visible. The global hit-position map of matched tracks (Fig. 8c) shows the same gaps as horizontal white bands at fixed ϕ, separately for $z < 0$ (east) and $z > 0$ (west). Cell occupancy (1–6) is uniform within a factor 1.224 (max/min).

Caveat

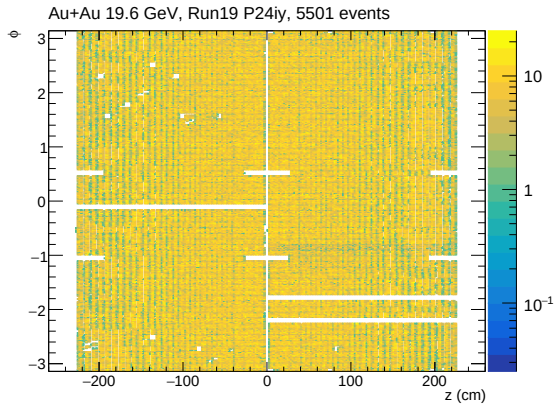
The dead-tray/module list is from one run only. A run-by-run (or time-period) BTOF status map is needed for the mixed-event step, since mixing events with different dead areas biases the random-match rate.



(a) Tray vs module for all stored BTOF hits (linear colour scale).



(b) Tray vs module of the cell matched to a QA track (flag>0), linear colour scale.



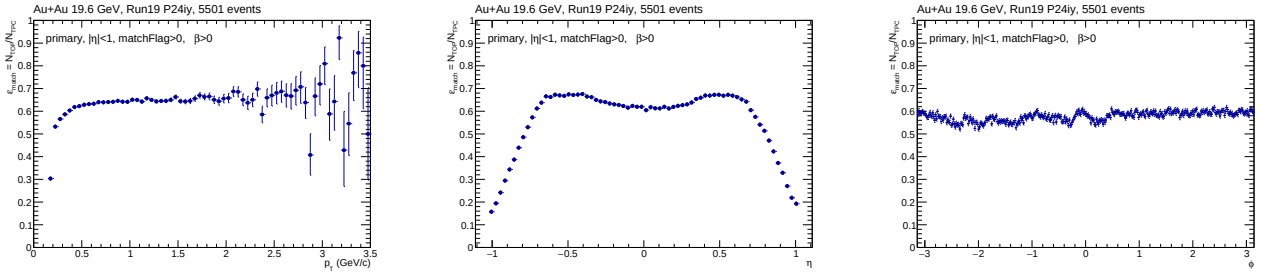
(c) Global ϕ vs z of the BTOF hit position of matched tracks, log colour scale.

Figure 8: BTOF occupancy.

Matching efficiency. Table 5 and Fig. 9a give $\varepsilon_{\text{match}}$ versus p_T. It rises from 0.467 at $0.15 < p_T < 0.3\,\text{GeV}/c$ to a plateau of 0.63–0.66 above $0.5\,\text{GeV}/c$ (Table 5). Versus η (Fig. 9b) it is 0.640 for $|\eta| < 0.5$ with a shallow dip at $\eta \approx 0$ and maxima near $|\eta| \approx 0.5$, then drops to 0.510 averaged over $0.5 < |\eta| < 1$ as the edge of the BTOF acceptance is reached. Versus ϕ (6° bins) it ranges from 0.539 (at $\phi \approx -2.36$) to 0.601.

Table 5: TOF matching efficiency vs p_T for QA tracks (generated table `tables/eff.tex`). Statistical errors.

p_T (GeV/c)	N_{TPC}	N_{TOF}	$\varepsilon_{\mathrm{match}}$
0.15–0.30	289 179	135 113	0.467 ± 0.001
0.30–0.50	289 953	175 653	0.606 ± 0.001
0.50–0.70	170 213	107 696	0.633 ± 0.001
0.70–1.00	126 415	81 087	0.641 ± 0.001
1.00–1.50	66 773	43 349	0.649 ± 0.002
1.50–2.00	14 077	9 177	0.652 ± 0.004
2.00–3.00	3 404	2 256	0.663 ± 0.008
3.00–5.00	141	98	0.695 ± 0.039



(a) $\varepsilon_{\mathrm{match}}$ vs p_T (0.05 GeV/c bins). (b) $\varepsilon_{\mathrm{match}}$ vs η. (c) $\varepsilon_{\mathrm{match}}$ vs ϕ (1° bins).

Figure 9: TOF matching efficiency $N_{\mathrm{TOF}}/N_{\mathrm{TPC}}$ (flag>0, $\beta > 0$ over QA tracks). Error bars: 68% Bayesian binomial intervals.

7 Mixed-event setup for TPC–BTOF matching

This section describes step 2: projecting every QA track to BTOF with the helix stored in the picoDst, matching it at the cell level to the BTOF hits of the *same* event (SE) and of *other* events (mixed events, ME), and validating the projection against the production matching. Unless stated otherwise all numbers are from the same 10k-event test file as Secs. 4–6 (5 501 selected events), produced by `StRoot/StTofMEMaker` and summarised by `analysis_note/scripts/make_me.C` (`logs/me_numbers_test.txt`).

7.1 Method

BTOF geometry from data. The picoDst does not contain the BTOF geometry, and the official geometry (`StBTofGeometry`) needs the full detector description plus database alignment. Instead, the position of every module is fitted from production-matched tracks. For these, `btofHitPos()` is the production projection point in global coordinates and (y_{local}, z_{local}, cell) are its local coordinates, so for each module

$$\vec{r}_{\mathrm{hit}} = \vec{C} + y_{\mathrm{local}}\,\vec{U} + z_{\mathrm{local}}\,\vec{V} + (\mathrm{cell} - 3.5)\,\vec{W}, \tag{2}$$

which is linear in the 12 parameters $(\vec{C}, \vec{U}, \vec{V}, \vec{W})$. The maker accumulates the normal-equation sums per module (`hGeomSum`, additive over jobs) and `macros/makeGeom.C` solves them (modules with ≥ 20 matched tracks). Of 3840 modules, 3699 are fitted, with a mean residual of 0.0029 cm, i.e. the picoDst position precision, so the linear model is exact. The fit gives $|\vec{U}| = 1.00000$ (mean over modules), a

cell pitch of 3.450 cm (RMS 0.001 cm over modules), the `mWidthPad` value of `StBTofMatchMaker`, and module-centre radii of 213.54 cm (RMS 0.78 cm; Fig. 10a). The remaining 141 modules (dead in this run or with too few tracks) are copied from the same module of the nearest fitted tray in the same half (east/west), rotated about z by the tray-to-tray ϕ difference. Applying the same rotation to fitted modules (closure test) reproduces their centres to 0.297 cm on average (RMS 0.219 cm, at most 1.07 cm; Fig. 10b), small compared with the 3.45 cm pitch. Only 3.70% of the projected tracks land in a rotated module.

Choice

Data-driven module geometry instead of `StBTofGeometry`: it includes the production alignment automatically, has no database dependence, and is validated to the 0.01 cm level by the fit residuals. Geometry file: `geom/btof_geom_test.txt` (from the test run 20093036).

Projection. For each QA track (Sec. 3.2) the global helix is built with `StPicoTrack::helix(B)` (origin = DCA point, global momentum, charge, B from the event). It is intersected with a cylinder of radius 213.5 cm (the mean module radius) to select candidate modules ($|\Delta\phi| < 0.09$ to the tray centre, $|\Delta z| < 12$ cm to the module centre); then the exact intersection with each candidate module plane gives the local coordinates, the cell (pitch 3.45 cm) and z_{local}. The projection is accepted if it falls inside one of the six cells and $|z_{\mathrm{local}}| < 3.05$ cm (the production window). If several modules qualify, the one with smallest $|z_{\mathrm{local}}|$ is kept.

Matching definitions. Hits are stored only as cell IDs, so all matches are at the cell level:

- **cell**: a BTOF hit in the projected cell;
- **cell±1 (production-like)**: a hit in the projected cell or in an adjacent cell of the same module. This is the production rule: `StBTofMatchMaker` (SL24y, "match find neighbours") accepts a hit in the projected cell or its ±1 neighbour (checked in the source).

P_{SE} is the fraction of projected tracks matched to a hit of the same event. P_{ME} is the fraction of (track, partner-event) trials in which the projected cell (or cell±1) has a hit in the partner event. $f_{\mathrm{rand}} = P_{\mathrm{ME}}/P_{\mathrm{SE}}$ estimates the fraction of SE matches that are random. To first order the true matching efficiency is $\varepsilon_{\mathrm{true}} = (P_{\mathrm{SE}} - P_{\mathrm{ME}})/(1 - P_{\mathrm{ME}})$.

Mixing. Partner events come from a FIFO buffer of the previous 5 selected events with the same run number, the same V_z class (14 classes of 10 cm) and the same refMult class (boundaries 0, 10, 25, 50, 80, 120, 170, 230, 300). Buffers are cleared at each run change, so dead areas are identical in mixed pairs. 88.5% of the events have the full 5 partners.

Choice

Mixing classes in refMult (not number of BTOF hits) and V_z, depth 5, same run only. refMult is independent of BTOF, and the hit occupancy follows it closely (Fig. 2a); P_{ME} is then studied directly versus the hit multiplicity of the partner event.

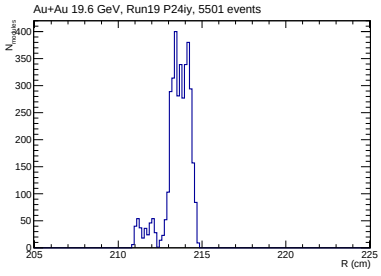
7.2 Validation of the projection

Of the 960 080 QA tracks, 76.8% have an accepted projection; the rest have no crossing with the TOF radius (1.8%), fall between modules (9.7%), outside the six cells (1.1%) or outside $|z_{\mathrm{local}}| < 3.05$ cm (10.5%) (Fig. 10c).

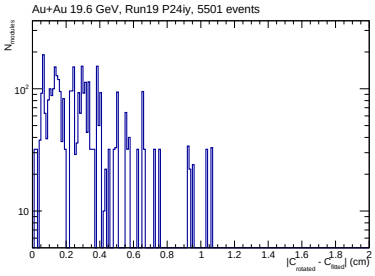
For the 562 288 production-matched QA tracks, 97.7% have an accepted projection, 92.9% are projected to the same module as production and 78.1% to the same cell (Fig. 10d). Above $p = 1.5\,\mathrm{GeV}/c$ the values are 98.2% (module) and 92.9% (cell); for $0.2 < p < 0.4\,\mathrm{GeV}/c$ the same-cell fraction is only 61.6%. Different cells in the same module are expected: 8.5% of the production matches have $|y_{\mathrm{local}}| > 1.75\,\mathrm{cm}$, i.e. production matched them to a hit in the neighbouring cell. When the cells agree, the local-coordinate differences (Figs. 10e, 10f) have RMS 0.043 cm (y) and 0.049 cm (z) for $p > 1.5\,\mathrm{GeV}/c$, growing to 0.679 cm and 0.454 cm for $p < 0.45\,\mathrm{GeV}/c$.

Caveat

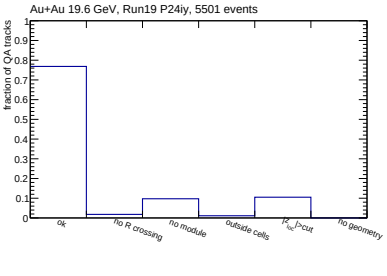
The picoDst helix starts at the DCA to the primary vertex, while production projects the outer TPC helix. Energy loss and multiple scattering between the TPC and BTOF are therefore not in this projection, which is why the agreement degrades at low momentum and P_{SE} is below the production match rate for $p_T < 0.3\,\mathrm{GeV}/c$ (Table 6). P_{ME} is not affected (Sec. 7.3).



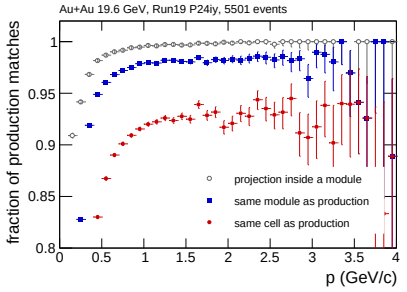
(a) Radius of the fitted module centres.



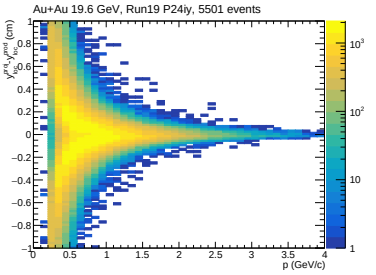
(b) Closure test of the rotated geometry: distance between rotated and fitted module centres, log scale.



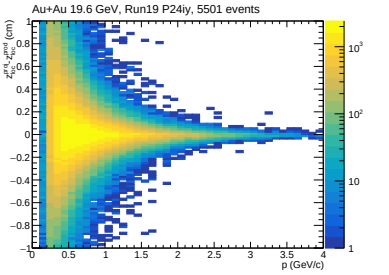
(c) Projection status of QA tracks (fractions).



(d) For production-matched tracks: fraction with an accepted projection (grey), projected to the production module (blue), and to the production cell (red), vs p.



(e) $y^{\mathrm{proj}}_{\mathrm{local}} - y^{\mathrm{prod}}_{\mathrm{local}}$ vs p when the cells agree, log colour scale.



(f) $z^{\mathrm{proj}}_{\mathrm{local}} - z^{\mathrm{prod}}_{\mathrm{local}}$ vs p when the cells agree, log colour scale.

Figure 10: Geometry and projection validation (test sample).

7.3 Same-event vs mixed-event matching (test sample)

737 720 QA tracks are projected and tested against 3 372 677 partner events in total. Integrated over p_T:

	P_{SE}	P_{ME} (%)	$f_{\mathrm{rand}} = P_{\mathrm{ME}}/P_{\mathrm{SE}}$ (%)
cell	0.682	2.34	3.4
cell±1 (production-like)	0.759	5.24	6.9

The production match rate for the same projected tracks is 0.734. The statistical uncertainty on P_{ME} is $\leq 0.01\%$ (absolute).

Table 6: Matching probabilities vs p_T for projected QA tracks (generated table `tables/me_eff_test.tex`). P_{prod}: production match with $\beta > 0$. Statistical errors on P_{ME}; those on P_{SE} and P_{prod} are below 0.003.

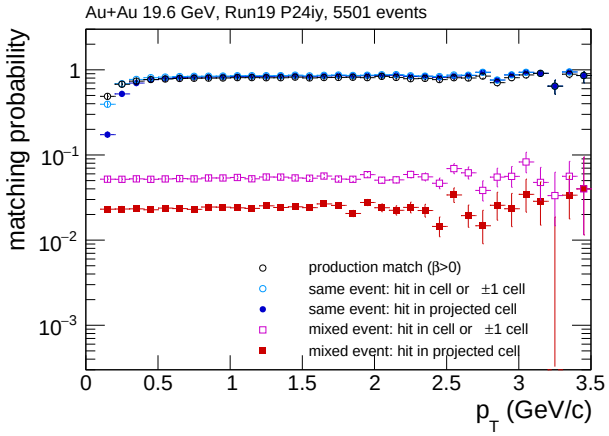
			projected cell			cell or ±1 (production-like)		
p_T (GeV/c)	N_{proj}	P_{prod}	P_{SE}	P_{ME} (%)	f_{rand} (%)	P_{SE}	P_{ME} (%)	f_{rand} (%)
0.15–0.30	203 994	0.625	0.427	2.316 ± 0.016	5.4	0.608	5.184 ± 0.023	8.5
0.30–0.50	229 351	0.751	0.729	2.304 ± 0.015	3.2	0.788	5.215 ± 0.022	6.6
0.50–0.70	135 791	0.785	0.800	2.365 ± 0.019	3.0	0.827	5.252 ± 0.028	6.3
0.70–1.00	101 000	0.798	0.823	2.380 ± 0.022	2.9	0.842	5.312 ± 0.033	6.3
1.00–1.50	53 358	0.810	0.838	2.420 ± 0.031	2.9	0.855	5.408 ± 0.046	6.3
1.50–2.00	11 319	0.810	0.844	2.489 ± 0.068	3.0	0.859	5.425 ± 0.100	6.3
2.00–3.00	2 789	0.808	0.850	2.292 ± 0.133	2.7	0.864	5.361 ± 0.200	6.2

Main observations:

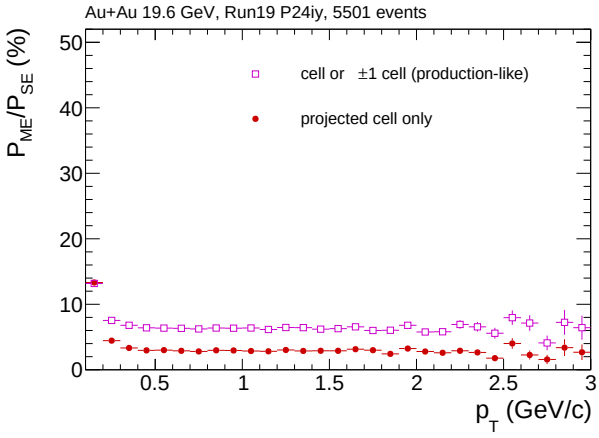
- P_{ME} **is the hit occupancy.** P_{ME} grows linearly with the number of hits N_{hits} in the partner event (Fig. 11c). A fit $P_{\mathrm{ME}} = a\, N_{\mathrm{hits}}$ gives $a = 4.31 \times 10^{-5}$, which is 0.99 times the uniform-occupancy expectation 1/23040. Averaged over trials, $P_{\mathrm{ME}}/(\langle N_{\mathrm{hits}} \rangle / 23040) = 0.99$.
- P_{ME} **is nearly flat** (2.3–2.5% for the cell definition) in p_T, η and in the local coordinates (Figs. 11a, 11e, 11f, 11g), as expected for hits unrelated to the track; it rises with refMult (Fig. 11d). Because random hits do not depend on the track, P_{ME} does not depend on the projection precision.
- **Random-match fraction.** For the production-like definition f_{rand} is 6.3% for $0.5 < p_T < 2\,\mathrm{GeV}/c$ and 8.5% for $0.15 < p_T < 0.3\,\mathrm{GeV}/c$ (Fig. 11b); the increase at low p_T comes from the lower P_{SE} there.
- **Same-event shape.** P_{SE} falls towards the cell edges in y_{local} (0.726 for $|y| < 1\,\mathrm{cm}$, 0.604 for $1.3 < |y| < 1.6\,\mathrm{cm}$) and at $|z_{\mathrm{local}}| \gtrsim 2.8\,\mathrm{cm}$; there the true hit is often in the neighbouring cell or module.
- First-order true matching efficiency: $\varepsilon_{\mathrm{true}} = 0.746$ (cell±1) and 0.674 (cell).

Caveat

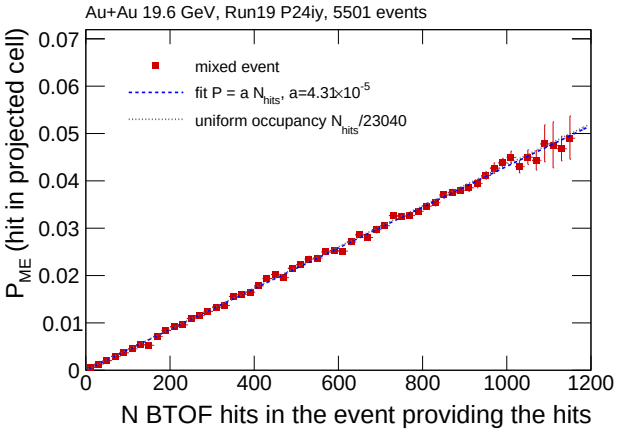
f_{rand} treats every projected track as equally exposed to random hits. Tracks whose own hit is present can still pick up a random hit in a neighbouring cell, which production would resolve differently. Production also resolves hits claimed by several tracks (flag 2) and requires a valid time, while P_{SE} here does not; this is why $P_{\mathrm{SE}}^{\pm 1}$ exceeds P_{prod} above $0.3\,\mathrm{GeV}/c$.



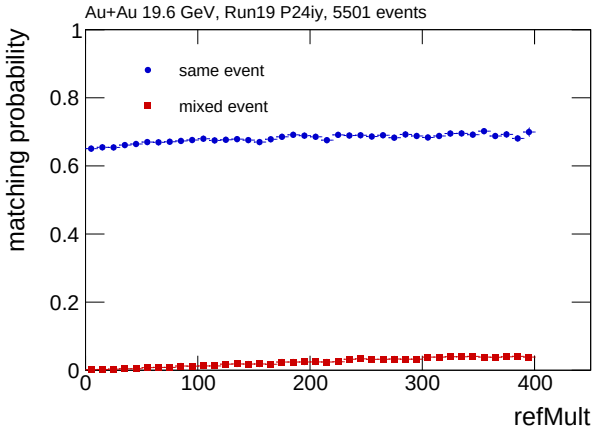
(a) Matching probabilities vs p_T (log scale): production (open black), SE cell±1 (open blue), SE cell (blue), ME cell±1 (open magenta), ME cell (red).



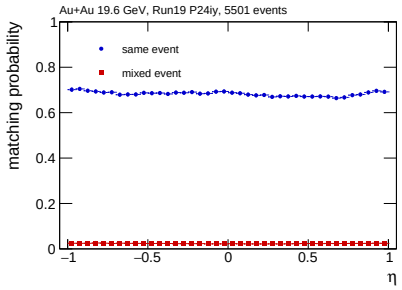
(b) $f_{\text{rand}} = P_{\text{ME}}/P_{\text{SE}}$ vs p_T (%) for cell±1 (open magenta) and cell (red).



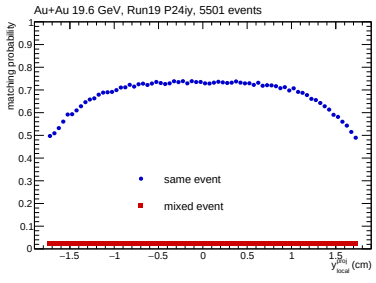
(c) P_{ME} (cell) vs number of BTOF hits in the partner event (red); dashed blue: linear fit through the origin; dotted grey: uniform occupancy $N_{\text{hits}}/23040$.



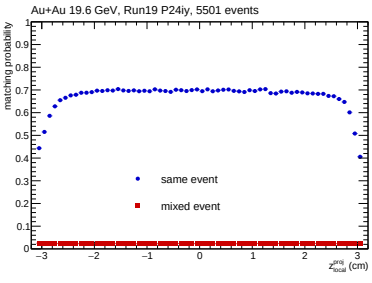
(d) P_{SE} (blue) and P_{ME} (red), cell definition, vs refMult of the track's event.



(e) P_{SE} and P_{ME} (cell) vs η.



(f) P_{SE} and P_{ME} (cell) vs projected y_{local}.



(g) P_{SE} and P_{ME} (cell) vs projected z_{local}.

Figure 11: Same-event (blue) and mixed-event (red) matching, test sample. Errors: 68% binomial intervals.

7.4 Mixed-event results with the 1M-event sample

Sample and processing. The large sample is 15 runs evenly spaced over the run range (20057003–20091008), one HTCondor job per run (`scripts/select_runs.py`, `scripts/me_v3.sub`, cluster 790125). Runs are taken only if they are not in the StRefMultCorr bad-run list for 2019 and if their largest files give at least 60 000 events. Files are deduplicated by name, because the catalog lists every replica as a separate entry (99 273 entries, 79 596 unique files; `logs/catalog_unique_P24iy.txt`). Files with a replica on the XRootD server qgp017 were excluded, because jobs reading from it stalled repeatedly. All 15 jobs finished normally (`scripts/merge_v3.sh`): 1 011 835 events read and 423 775 selected. With one run per job the 126 mixing buffers fill once per job, and 97.8% of the selected events have all 5 mixing partners. The jobs use the geometry fitted on the earlier large pass (`geom/btof_geom_1M.txt`).

Caveat

An earlier 1M-event pass (one file per job, draft v0.3 of this note) processed 5 files twice because of the duplicate catalog entries. It is superseded by this sample. Its integrated P_{ME} values (2.33% and 5.21%) agree with the ones below within 0.03 percentage points.

Caveat

The selected-event fraction is 42% here, against 55% in the test file, because the V_z distribution differs between runs. All rates below are per projected track or per mixed trial, so they are not affected, but event-level averages are not directly comparable with Sec. 4.

Geometry stability. The geometry refitted from the merged sums of the 15 jobs (`geom/btof_geom_v3.txt`) has 3697 fitted modules, fit residual 0.0029 cm and pitch 3.450 cm. For the 3696 modules fitted in both, the module centres differ from the test-run geometry by 0.0005 cm on average (at most 0.006 cm). The BTOF geometry is stable across the runs used, and the test-run geometry is adequate.

Projection and matching. With 71 105 376 QA tracks and 268 305 132 mixed trials, the projection-level numbers of the test sample are reproduced: 76.8% of the tracks are projected, and for production-matched tracks 92.8% (cell) and 98.2% (module) agree above 1.5 GeV/c. The integrated probabilities are

	P_{SE}	P_{ME} (%)	$f_{\mathrm{rand}} = P_{\mathrm{ME}}/P_{\mathrm{SE}}$ (%)
cell	0.679	2.34	3.4
cell±1 (production-like)	0.757	5.24	6.9

with statistical uncertainties below 10^{-4}, compared with 2.34% and 5.24% in the test sample. f_{rand} (cell±1) is 6.3% for $0.5 < p_T < 2$ GeV/c and 8.5% for $0.15 < p_T < 0.3$ GeV/c (Table 7, Fig. 12b). P_{ME} equals the trial-averaged hit occupancy ($P_{\mathrm{ME}}/(\langle N_{\mathrm{hits}}\rangle/23040) = 1.00$; slope 1.00 times 1/23040). These are probabilities that a random hit is *present* in the matching cells. How often such a hit is actually chosen instead of the track's own hit, and which m^2 it then gives, is the subject of Sec. 7.5.

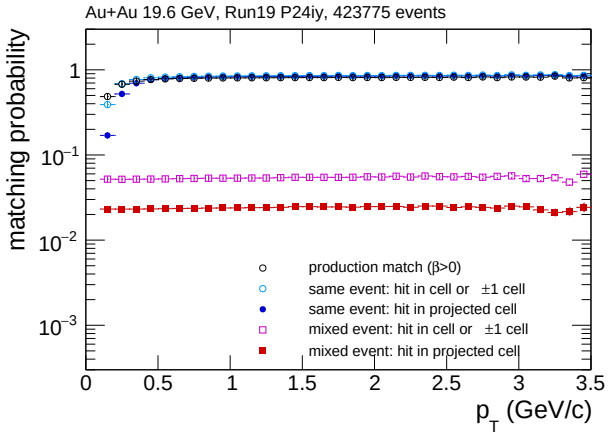
Caveat

P_{ME} is not exactly flat in p_T (Table 7). The origin is not understood. Candidates: harder tracks come more often from higher-refMult events within a mixing class, or point preferentially to higher-occupancy regions. This was not checked.

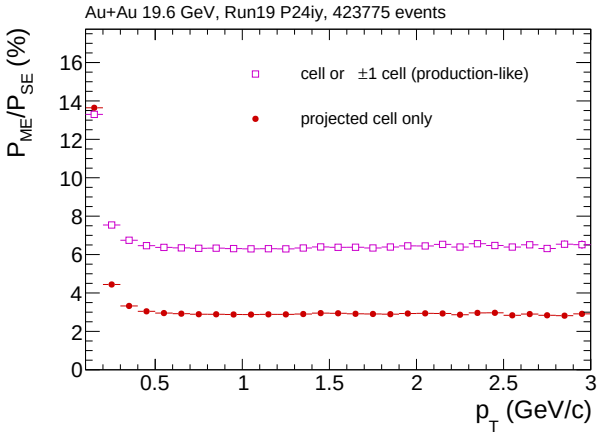
Table 7: Matching probabilities vs p_T, 1M-event sample (`tables/me_eff_v3.tex`); notation as in Table 6.

			projected cell			cell or ±1 (production-like)		
p_T (GeV/c)	N_{proj}	P_{prod}	P_{SE}	P_{ME} (%)	f_{rand} (%)	P_{SE}	P_{ME} (%)	f_{rand} (%)
0.15–0.30	15 119 485	0.625	0.426	2.314 ± 0.002	5.4	0.607	5.184 ± 0.003	8.5
0.30–0.50	17 034 547	0.751	0.726	2.319 ± 0.002	3.2	0.786	5.201 ± 0.002	6.6
0.50–0.70	10 030 992	0.785	0.798	2.346 ± 0.002	2.9	0.825	5.249 ± 0.003	6.4
0.70–1.00	7 454 599	0.798	0.822	2.373 ± 0.003	2.9	0.841	5.319 ± 0.004	6.3
1.00–1.50	3 943 131	0.806	0.834	2.411 ± 0.003	2.9	0.851	5.372 ± 0.005	6.3
1.50–2.00	826 470	0.812	0.842	2.459 ± 0.008	2.9	0.857	5.471 ± 0.011	6.4
2.00–3.00	195 367	0.814	0.847	2.472 ± 0.016	2.9	0.860	5.563 ± 0.023	6.5

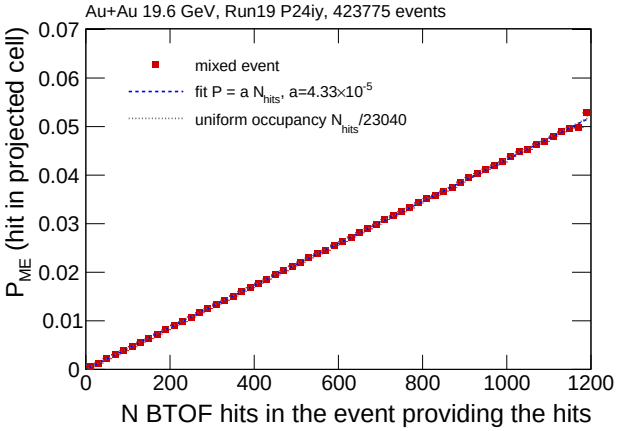
Run dependence. Per run, P_{ME} (cell±1) ranges from 5.10% to 5.38% (Fig. 12e), following the occupancy: the ratio $P_{\mathrm{ME}}/(\langle N_{\mathrm{hits}}\rangle/23040)$ stays between 0.99 and 1.01 (Fig. 12f). Per-run values: `logs/me_runs_v3.txt` (`scripts/make_me_runs.C`).



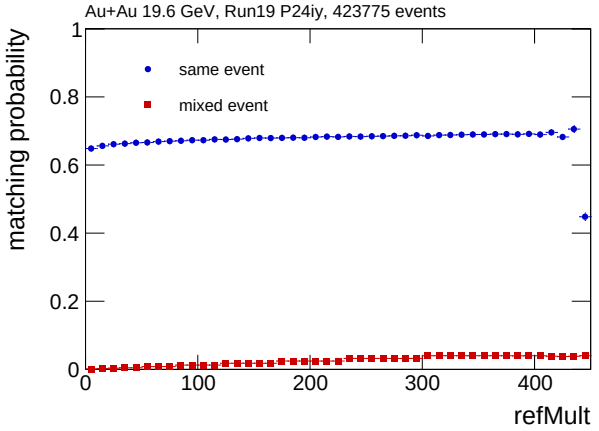
(a) Matching probabilities vs p_T, 1M sample; markers as in Fig. 11a.



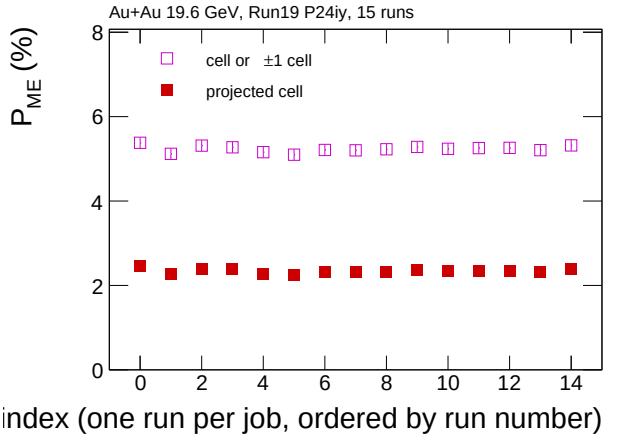
(b) $f_{\rm rand}$ vs p_T, 1M sample: cell±1 (open magenta), cell (red).



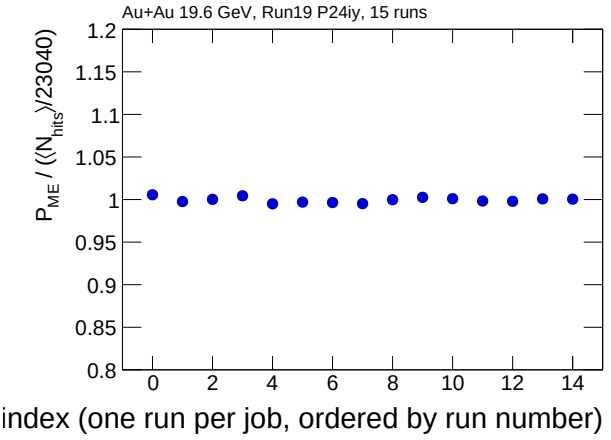
(c) $P_{\rm ME}$ (cell) vs number of BTOF hits in the partner event, with linear fit (dashed) and uniform occupancy (dotted).



(d) $P_{\rm SE}$ (blue) and $P_{\rm ME}$ (red) vs refMult, cell definition.

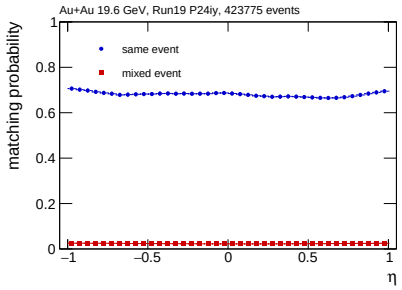


(e) $P_{\rm ME}$ per run (one job per run, ordered by run number): cell±1 (open magenta) and cell (red); statistical errors are smaller than the markers.

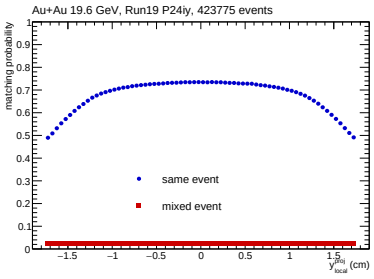


(f) $P_{\rm ME}$ (cell) per run divided by the trial-averaged occupancy $\langle N_{\rm hits}\rangle/23040$.

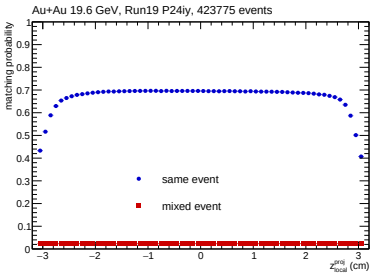
Figure 12: Mixed-event results, 1M sample (423 775 selected events from 15 runs).



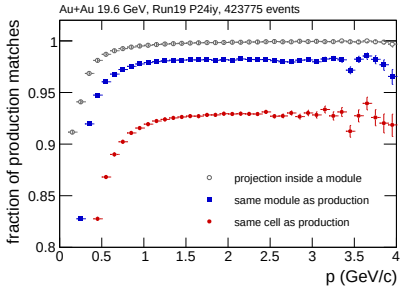
(a) P_{SE}, P_{ME} (cell) vs η.



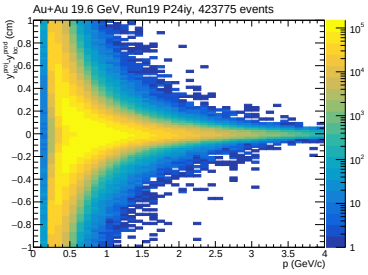
(b) P_{SE}, P_{ME} (cell) vs projected y_{local}.



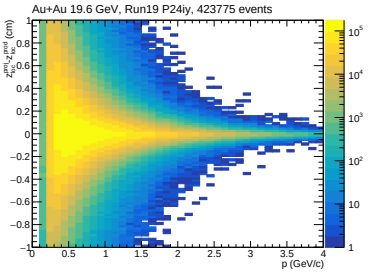
(c) P_{SE}, P_{ME} (cell) vs projected z_{local}.



(d) Projection vs production (as Fig. 10d).



(e) $\Delta y_{\mathrm{local}}$ vs p, same cell.



(f) $\Delta z_{\mathrm{local}}$ vs p, same cell.

Figure 13: Differential mixed-event and projection plots, 1M sample (same blue/red convention as Sec. 7.3).

7.5 m^2 of same-event and mixed-event matches per centrality

This section uses the 1M-event sample of Sec. 7.4 (`StRoot/StTofMEMaker` version 3, `output/me_v3.root`). It answers the question: what does the m^2 distribution of random TPC–BTOF matches look like, and how much of the measured m^2 distribution does it account for, in each momentum bin and centrality class? Numbers: `logs/m2_numbers_v3.txt` from `scripts/make_m2.C`.

7.5.1 Method

Centrality. Centrality is taken from StRefMultCorr (Au+Au 19.6 GeV 2019 parameters: V_z- and luminosity-corrected refMult, nine classes from 0–5% to 70–80%). Of the 423775 selected events, 873 are flagged as pile-up by its refMult–nBTOFMatch cut, 40647 are beyond 80%, and 497 have no BTOF hit with a time (no valid start time). The remaining 381758 events enter the m^2 histograms. Bad runs are excluded at the run selection. The mixing classes are unchanged (raw refMult).

Hit times. A picoDst BTOF hit carries only its cell. A time of flight exists for a hit only if a picoDst track was matched to it in production (`StPicoBTofPidTraits::btof()`, with $\beta > 0$). On average 35.3% of the hits per event have a time. Each cell therefore has one of three states: no hit, hit with time t, hit without time.

β and m^2. For each projected QA track the hit is chosen as in production: the projected cell, otherwise the nearer, then the farther neighbour, preferring hits that have a time (Sec. 7). The path length is $L =$ `tofPathLength`(primary vertex, projected point, helix curvature), the same function as

`StBTofMatchMaker`, and

$$\beta = \frac{L}{c\,t}, \qquad m^2 = p^2\left(\frac{1}{\beta^2} - 1\right), \tag{3}$$

with p the primary momentum. For tracks whose own production hit is chosen, L agrees with production ($L - \beta_{\mathrm{prod}} c\, t_{\mathrm{prod}}$: mean 0.001 cm, RMS 0.26 cm) and m^2 differs from the production value by $0.0007\,\mathrm{GeV}^2/c^4$ RMS. The same-event (SE) m^2 is therefore the production m^2 of the projected tracks.

Mixed event with overlay. A random hit only produces a wrong m^2 when it is chosen *instead of* the track's own hit. The mixed event (ME) is therefore built as an overlay: the hits of a partner event (same run, V_z and refMult class, up to 5 partners) are added on top of the hits of the track's own event, and the hit choice above is redone. When a partner hit is chosen (projected cell free in this event, or same cell with an earlier partner time), its time is combined with the track's L and p to give an m^2. Without the overlay, every partner hit present in the matching cells would count. That version (kept for reference, `hM2ME`) exceeds the SE yield in the π–K and K–p valleys by a factor 3.8–5.5 in 0–5% ($0.4 < p < 1.5\,\mathrm{GeV}/c$), which is impossible. The overlay keeps 34% of those entries.

Normalization. No fit is used. In each centrality class c and fine momentum bin j ($0.05\,\mathrm{GeV}/c$)

$$\frac{dN_{\mathrm{rand}}}{dm^2}(c,j) = \frac{dN_{\mathrm{ME}}}{dm^2}(c,j)\,\frac{N_{\mathrm{proj}}(c,j)}{N_{\mathrm{trials}}(c,j)}\,\frac{n_{\mathrm{t}} + n_{\mathrm{u}}}{n_{\mathrm{t}}}, \tag{4}$$

where N_{proj} counts projected tracks, N_{trials} the (track, partner) pairs, and n_{t} (n_{u}) the trials whose chosen partner hit has (has no) time. The last factor, 2.32 on average (2.41 in 70–80%, 2.32 in 0–5%), restores the random hits that have no stored time: only 43.2% of the partner hits chosen in the mixed events have a time. Coarse momentum bins are sums of the normalized fine bins.

Choice

All BTOF hits are included in the random-match estimate (user's decision). Random hits without a stored time are assumed to have the same time distribution as random hits with a time.

Caveat

Hits without a time are those no picoDst track was matched to: secondaries, particles not reconstructed in the TPC, noise. Their times are probably later on average than those of matched hits, so the shape of the random m^2 distribution may be biased towards small m^2; its normalization would be lower by a factor 2.32 if only timed hits were counted.

7.5.2 Results

Caveat

The overlay used here always keeps the track's own hit in the projected cell. Production instead picks the hit with the largest ToT among the projected and neighbouring cells, so the random matches below are underestimated. Sec. 7.6 studies this and the late background the picoDst cannot describe.

Figures 15a–18a overlay the normalized ME distribution on the SE m^2 distribution in nine momentum bins for 0–5%, 30–40%, 70–80% and 0–80%; the other centralities are in Appendix A. Table 8 and Fig. 14a give the random fraction under the π, K and p peaks.

- **Size.** Integrated over m^2 and $0.2 < p < 3\,\mathrm{GeV}/c$, random matches are 2.5% of the SE matches in 0–80%, 4.2% in 0–5%, 1.3% in 30–40% and 0.2% in 70–80%. This is smaller than f_{rand} of Sec. 7.4, because the track's own hit is usually chosen first.

- **Under the peaks (0–5%).** At $0.2 < p < 0.4\,\mathrm{GeV}/c$ the random fraction is 6.9% (π), 3.9% (K) and 3.0% (p); at $0.8 < p < 1.0\,\mathrm{GeV}/c$ it is 2.1%, 3.0% and 1.0%. In 30–40% the values are about three times smaller (0.6%, 0.9%, 0.3% at $0.8 < p < 1.0\,\mathrm{GeV}/c$), and in 70–80% they are below 0.4%. The kaon fraction stays roughly flat with p, because random m^2 values pile up between the pion and proton peaks.

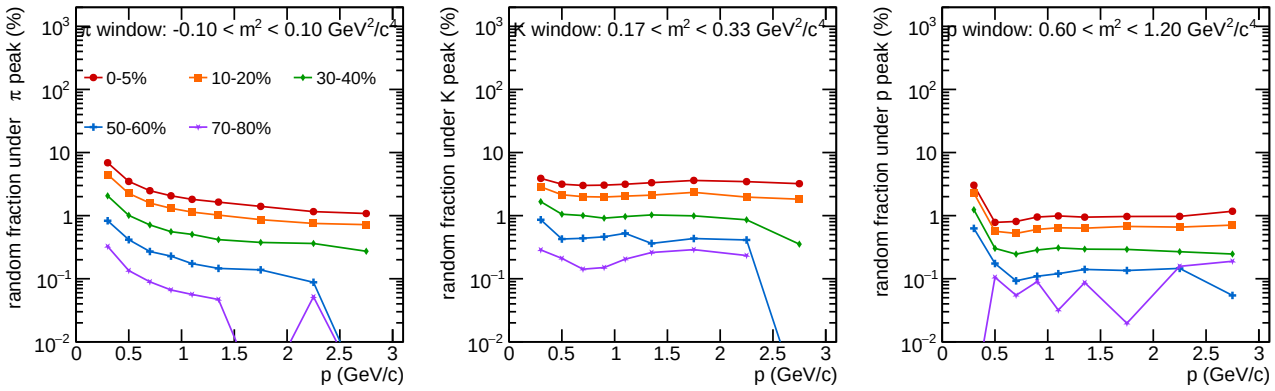
- **Shape (read by eye).** The random distribution peaks just above $m^2 = 0$ and falls steadily towards large m^2. At low momentum it follows the SE distribution in the π–K valley; at higher momentum it lies well below the SE distribution everywhere except in the valleys.

- **Consistency.** In 0–5%, $0.4 < p < 1.5\,\mathrm{GeV}/c$, the largest ME/SE ratio in any m^2 bin ($0.04\,\mathrm{GeV}^2/c^4$, at least 50 SE counts) is 1.07. In the lowest momentum bin (0.2–0.4 GeV/c) it is 1.21 (Table 8), so the estimate is somewhat too large there. The excess probably comes from the all-hits scaling, or from the less precise projection at low momentum (Sec. 7).

- **Tail above the proton.** For $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$ (between p and d) the SE yield is 7.41 times the random estimate in 0–80% (2–9 times in 0–5% above 0.6 GeV/c, Table 8). Random matches account for only a small part of this tail.

Caveat

Most of the SE yield between the proton and deuteron peaks is not from random hits. Candidates: decays in flight (the track's p does not belong to the particle that fired the cell), wrong primary vertex or out-of-time pile-up, and wrongly matched neighbour hits of the same event. Not studied.

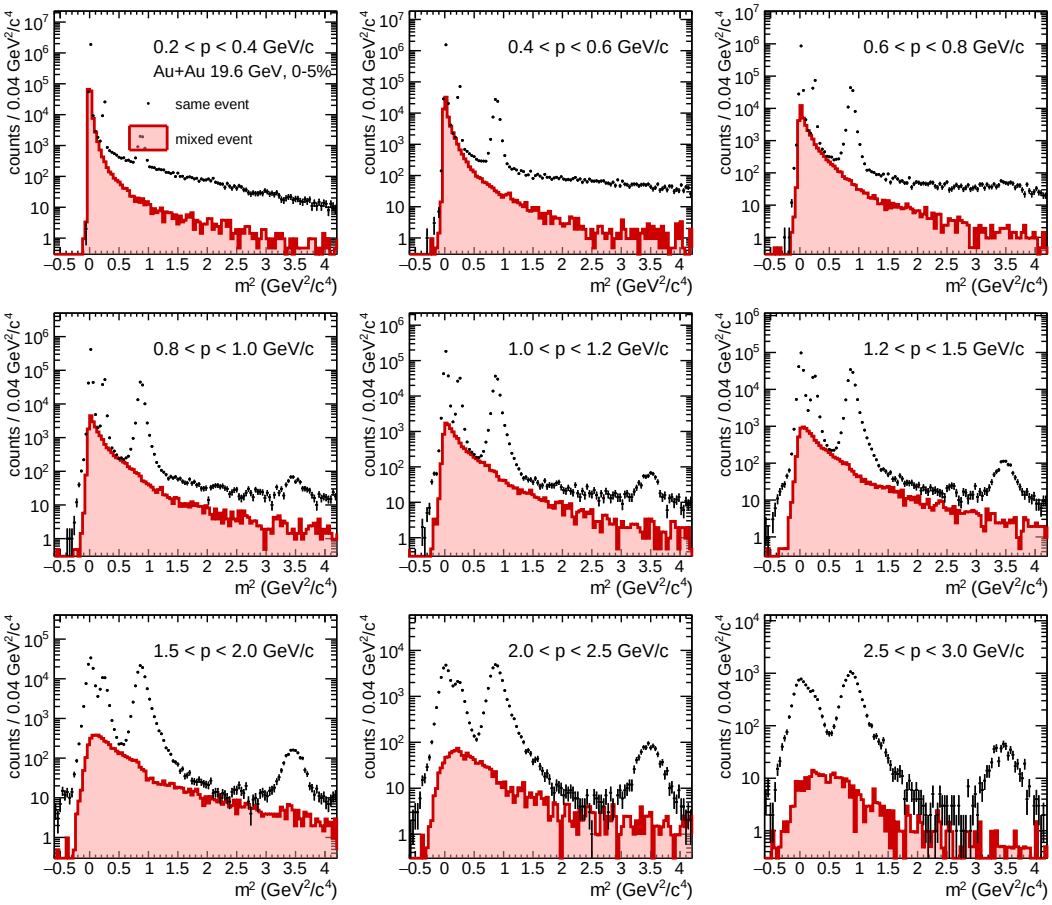


(a) Random-match fraction under the π (left), K (middle) and p (right) peaks vs momentum for 0–5% (red), 10–20% (orange), 30–40% (green), 50–60% (blue) and 70–80% (violet). Points with fewer than 100 SE counts in the window are omitted. ME: overlay, all hits.

Figure 14: Random fraction under the PID peaks.

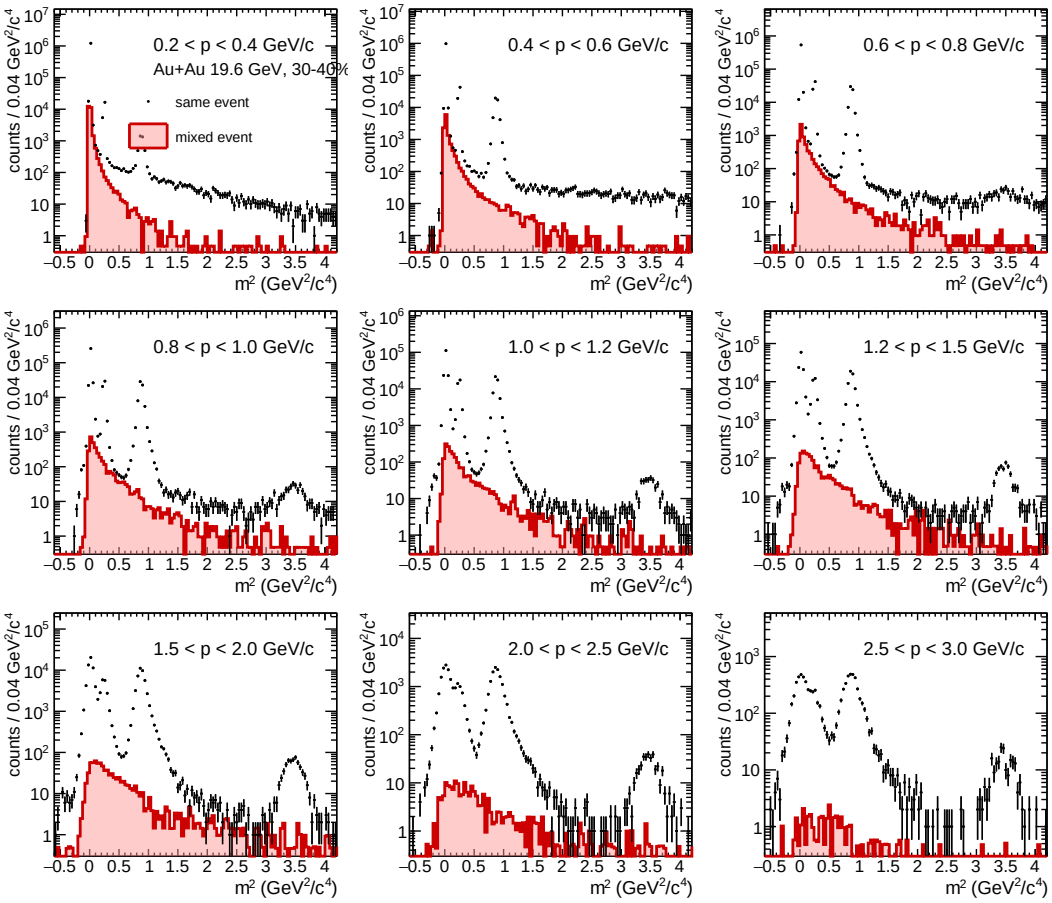
Table 8: Random-match fraction (%) of the SE yield in the π ($-0.1 < m^2 < 0.1$), K ($0.17 < m^2 < 0.33$) and p ($0.6 < m^2 < 1.2\,\mathrm{GeV}^2/c^4$) windows, SE/ME ratio between the p and d peaks, and the largest ME/SE ratio over m^2 bins with ≥ 50 SE counts in $-0.3 < m^2 < 3$. ME: overlay, all hits. Statistical errors not shown; they are large in 70–80% above 1 GeV/c. Generated: `tables/m2_contam_v3.tex`.

		random fraction under peak (%)			SE/ME	max
centrality	p (GeV/c)	π	K	p	$1.6 < m^2 < 2.8$	ME/SE
0–5%	0.2–0.4	6.88	3.89	3.03	22.4	1.21
	0.4–0.6	3.49	3.15	0.78	25.6	1.07
	0.6–0.8	2.48	3.01	0.81	8.6	0.98
	0.8–1.0	2.06	3.04	0.95	5.0	1.04
	1.0–1.2	1.81	3.14	0.99	3.7	0.91
	1.2–1.5	1.64	3.33	0.95	2.6	0.82
	1.5–2.0	1.40	3.61	0.97	2.1	0.59
	2.0–2.5	1.16	3.46	0.97	2.9	0.31
	2.5–3.0	1.08	3.21	1.17	4.9	0.16
30–40%	0.2–0.4	2.06	1.66	1.24	65.8	0.81
	0.4–0.6	1.01	1.05	0.30	28.8	0.82
	0.6–0.8	0.71	1.00	0.25	7.9	0.70
	0.8–1.0	0.56	0.91	0.29	6.6	0.70
	1.0–1.2	0.50	0.97	0.31	3.9	0.53
	1.2–1.5	0.42	1.03	0.29	3.5	0.56
	1.5–2.0	0.38	0.99	0.29	3.0	0.27
	2.0–2.5	0.36	0.86	0.27	5.4	0.13
	2.5–3.0	0.27	0.35	0.25	23.5	0.03
70–80%	0.2–0.4	0.33	0.29	0.00	111.3	0.17
	0.4–0.6	0.13	0.21	0.11	71.4	0.09
	0.6–0.8	0.09	0.14	0.05	24.7	0.06
	0.8–1.0	0.07	0.15	0.09	0.0	0.01
	1.0–1.2	0.06	0.20	0.03	6.7	0.04
	1.2–1.5	0.05	0.26	0.09	0.0	0.01
	1.5–2.0	0.00	0.29	0.02	0.0	0.01
	2.0–2.5	0.05	0.23	0.16	6.3	0.01
	2.5–3.0	0.00	0.00	0.19	0.0	0.00



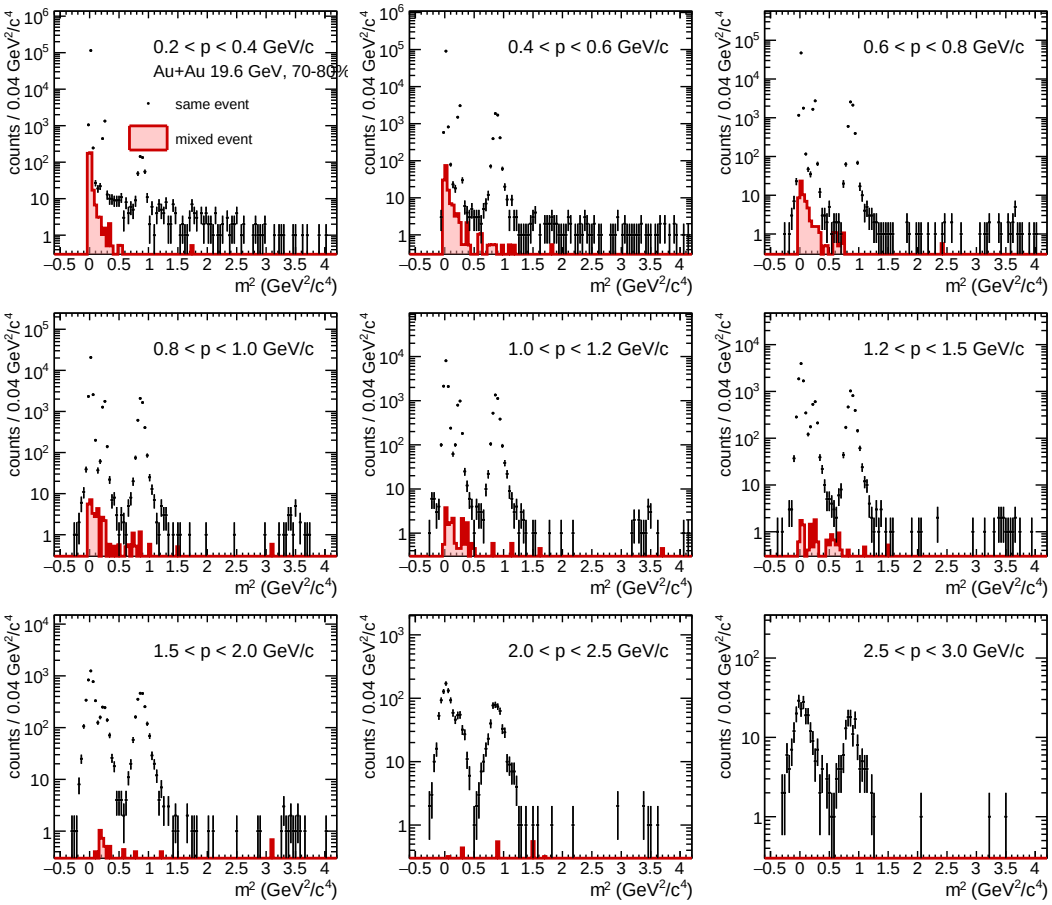
(a) m^2 of projected QA tracks, 0–5%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 15: SE vs ME m^2, 0–5%.



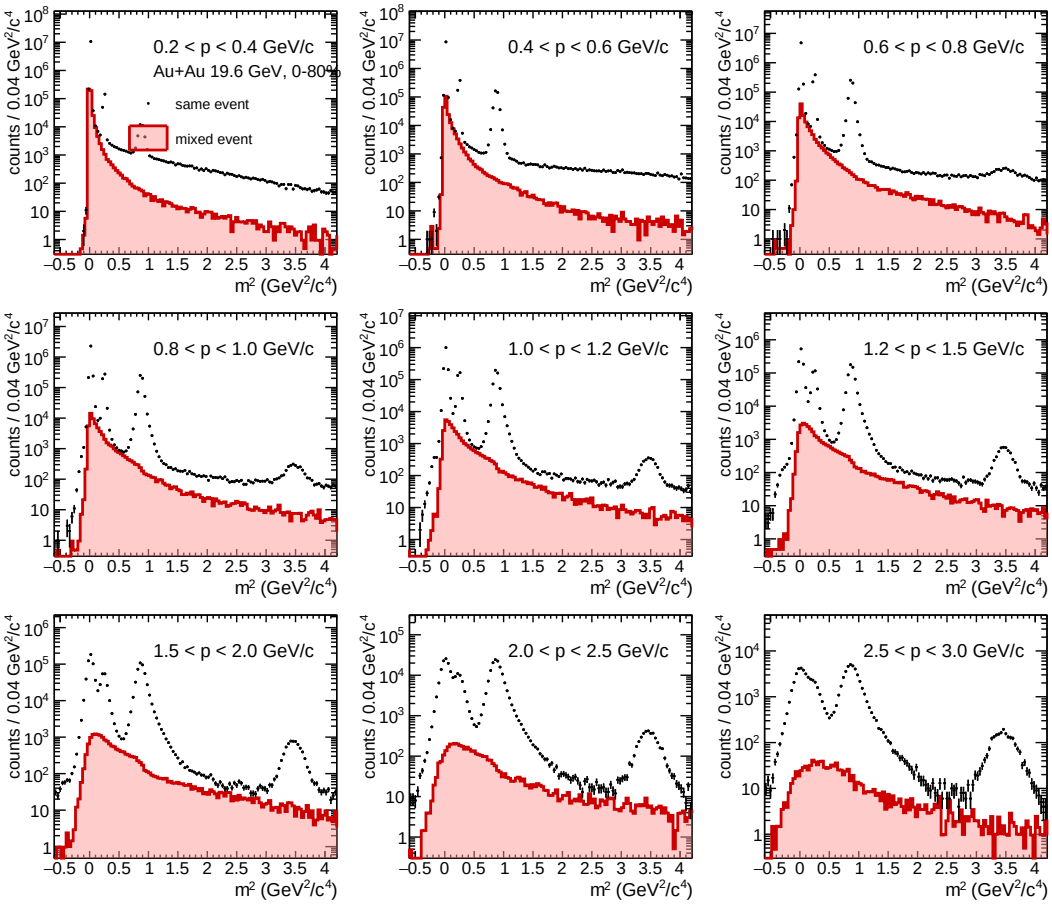
(a) m^2 of projected QA tracks, 30–40%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 16: SE vs ME m^2, 30–40%.



(a) m^2 of projected QA tracks, 70–80%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 17: SE vs ME m^2, 70–80%.



(a) m^2 of projected QA tracks, 0–80% (sum of the nine classes), nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 18: SE vs ME m^2, 0–80%.

7.6 Origin of the same-event background in m^2

Sec. 7.5 found that the overlay mixed event does not describe the same-event (SE) m^2 away from the peaks. In particular, the SE yield between the proton and deuteron peaks ($1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$, called HI below) is several times larger than the mixed-event (ME) estimate. This section reports the study of that mismatch. It uses four extra passes of `StRoot/StTofMEMaker` over the same 15 runs and events as Sec. 7.4:

- v4: SE and ME split by 12 track, event and hit variables;
- v5: the same quantities per projected BTOF cell;
- v6: details of neighbour-cell matches, on the first 25 000 events of each run;
- v7, v8: two emulations of the production hit choice.

Scripts: `scripts/make_bkg.C`, `make_cell.C`, `make_nb.C`, `make_m2.C`, `make_late.C`; logs `logs/bkg_v4.txt`, `cell_v5.txt`, `nb_v6.txt`, `m2_numbers_v7.txt`, `m2_numbers_v8.txt`, `late_v8.txt`. All comparisons are for 0–80% centrality unless stated otherwise.

7.6.1 Where the HI excess is

Table 9 lists the HI yield per 10^4 projected tracks for SE and for the overlay ME of Sec. 7.5, split by each variable.

- **No dependence:** the SE/ME ratio does not depend on the difference between the TPC and VPD vertex (≈ 6 in every class, including events without a VPD vertex), on $|z_{\mathrm{local}}|$, on η or on the track charge. This rules out start-time errors and event pile-up as the main source (Fig. 19b, 19c).
- **Mild dependence:** the ratio rises with the DCA ($5.9 \rightarrow 9.0$) and is larger for short tracks. Tracks with $n\sigma_{\mathrm{p}} > 8$ (light-nucleus candidates) show a strong excess, but they are a small share.
- **Strong dependence, the chosen cell:** the HI excess comes from tracks matched to a hit in a *neighbouring* cell (Fig. 19a). Neighbour-cell matches are 6% of the SE matches but 82% of the SE HI entries (v6). SE/ME in the HI window is 10.5 for them and 2.0 for projected-cell matches.

The neighbour-cell excess is not caused by dead or inefficient cells. Cells with an efficiency above 0.8 (79% of the projected tracks) carry 70%+13% of the HI entries, and cells below 0.5 carry about 1% (Table 10, Fig. 19d). It is not a projection effect either. The rate of HI neighbour matches is about 7×10^{-4} per SE match whether the track crosses the centre or the edge of the projected cell (Fig. 20a). At the centre the neighbour cell is $\geq 1.7\,\mathrm{cm}$ away, against a projection precision of about $0.05\,\mathrm{cm}$ at these momenta. Finally, in 219063 of the 350881 neighbour matches (v6, $0.6 < p < 1.5\,\mathrm{GeV}/c$) the projected cell *did* contain a hit, but that hit has no time in the picoDst: production did not match it to any track and gave our track the neighbour hit instead. The HI fraction of these matches is 15.3 per mille of the peak yield, against 6.1 per mille when the projected cell was empty.

Table 9: HI yield ($1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$) per 10^4 projected tracks for SE and overlay ME (projected cell first, all hits), by category. "share" is the category's share of all SE matches. *Hit variables: absolute counts instead of rates, because projected tracks cannot be assigned to them. Generated: `tables/bkg_cat_v4.tex`.

		$0.6 < p < 1.5\,\mathrm{GeV}/c$			$1.5 < p < 3\,\mathrm{GeV}/c$			
variable	category	SE	ME	SE/ME	SE	ME	SE/ME	share
gDCA	<0.5 cm	6.94	1.18	5.9	15.91	5.15	3.1	75.9%
	0.5-1 cm	7.29	1.13	6.4	23.89	5.46	4.4	15.7%
	1-2 cm	8.65	1.23	7.1	59.15	5.32	11.1	6.1%
	2-3 cm	10.92	1.22	9.0	175.75	7.43	23.7	2.3%
$\lvert V_z - V_z^{\mathrm{VPD}}\rvert$	<3 cm	6.70	1.11	6.0	19.36	4.89	4.0	51.7%
	3-6 cm	6.93	1.15	6.0	18.80	5.32	3.5	22.1%
	>6 cm	7.83	1.30	6.0	21.13	5.41	3.9	23.0%
	no VPD	9.61	1.54	6.2	20.02	7.72	2.6	3.2%
$\lvert z_{\mathrm{local}}\rvert$	<1 cm	7.11	1.04	6.8	19.65	4.75	4.1	36.1%
	1-2 cm	7.15	1.08	6.6	19.55	4.86	4.0	35.4%
	2-2.6 cm	7.01	1.14	6.2	19.43	4.82	4.0	19.5%
	>2.6 cm	7.12	1.95	3.6	20.49	8.22	2.5	8.9%
$\lvert y_{\mathrm{local}}\rvert$	<0.6 cm	6.51	0.78	8.4	17.51	3.32	5.3	36.0%
	0.6-1.2 cm	6.86	0.82	8.3	18.73	3.67	5.1	35.5%
	>1.2 cm	8.06	2.10	3.8	23.22	9.43	2.5	28.5%
nHitsFit	15-24	14.03	5.14	2.7	24.75	9.11	2.7	0.2%
	25-34	8.57	1.84	4.7	32.52	5.58	5.8	2.0%
	≥35	7.01	1.13	6.2	19.05	5.11	3.7	97.8%
$\lvert\eta\rvert$	<0.3	6.49	1.15	5.6	24.03	5.41	4.4	33.1%
	0.3-0.6	7.24	1.19	6.1	20.15	5.24	3.8	35.1%
	>0.6	7.43	1.16	6.4	17.89	5.04	3.6	31.8%
centrality	0-10%	9.83	1.70	5.8	24.52	7.55	3.2	34.2%
	10-40%	6.23	1.03	6.1	17.65	4.34	4.1	52.2%
	40-80%	3.13	0.35	8.8	12.49	1.11	11.3	13.6%
hit cell	projected	2378.00	1168.69	2.0	1437.00	537.28	2.7	84.9%*
	neighbour	10912.00	1034.81	10.5	2417.00	481.07	5.0	15.1%*
hit owner	own	13180.00	0.00	0.0	3830.00	0.00	0.0	99.7%*
	own, shared	0.00	0.00	0.0	0.00	0.00	0.0	0.0%*
	other track	110.00	2203.49	0.0	24.00	1018.34	0.0	0.3%*
	other, shared	0.00	0.00	0.0	0.00	0.00	0.0	0.0%*
hits in cell	1	13290.00	2203.49	6.0	3854.00	1018.34	3.8	100.0%*
	≥2	0.00	0.00	0.0	0.00	0.00	0.0	0.0%*
$n\sigma_{\mathrm{p}}$	<-3	6.61	1.08	6.1	34.77	6.57	5.3	85.4%
	-3..3	8.21	1.63	5.0	18.93	5.03	3.8	14.2%
	3..8	12.45	1.51	8.2	26.05	9.81	2.7	0.3%
	>8	117.38	5.91	19.8	109.97	15.90	6.9	0.0%
charge	-	6.96	1.22	5.7	17.19	5.53	3.1	45.9%
	+	7.21	1.16	6.2	20.82	5.13	4.1	54.1%

Table 10: HI yield by cell efficiency ε_c (fraction of tracks projected to the cell for which the cell has a hit), $0.6 < p < 3\,\mathrm{GeV}/c$. "nb. match": share of the projected tracks matched to a neighbour cell. Generated: `tables/cell_eff_v5.tex`.

ε_c	cells	tracks (%)	nb. match (%)	nb. cell HI (10^{-4})		proj. cell HI (10^{-4})		share of HI (%)
				SE	ME	SE	ME	
0.0–0.2	935	4.1	0.5	1.08	0.35	0.24	0.00	0.6
0.2–0.5	77	0.3	4.2	10.97	2.35	5.09	2.16	0.6
0.5–0.8	3636	16.2	4.6	5.23	0.93	2.84	1.04	15.8
0.8–0.9	15284	66.8	5.6	6.87	0.74	1.79	0.86	69.8
0.9–1.0	3108	12.6	5.3	7.39	0.59	1.31	0.70	13.2

7.6.2 Production chooses among several hits by ToT

In `StBTofMatchMaker` (SL24y, lines 782–833) a track with hits in more than one candidate cell (projected cell and neighbours) is matched to the hit with the *largest ToT* (match flag 2). Only for equal ToT is the hit closest in y used (flag 3). So a random hit in a neighbour cell can take the place of the track's own hit. The overlay ME of Sec. 7.5 always kept the own hit in the projected cell and therefore misses these cases. The picoDst has no ToT, so two emulations were run:

- **Equal chance** (v7): ToT independent of the hit's origin. Each of the hits in the candidate cells, from this event and from the partner, wins with probability $1/(n_A + n_B)$. Untimed partner hits are included through the scaling of Sec. 7.5.
- **Earliest time wins** (v8; ToT replaced by the time, smaller time wins): the hit chosen for the track in this event competes with the timed partner hits in the candidate cells. Untimed partner hits cannot take part.

	proj. cell first	equal chance	earliest time
random matches / SE matches, 0–80% (%)	2.5	4.3	1.9
0–5% (%)	4.2	7.3	3.3
70–80% (%)	0.2	0.3	0.1
max ME/SE per m^2 bin, 0–5%, $0.4 < p < 1.5$	1.07	3.08	0.83
SE/ME, HI window, 0–80%, $0.2 < p < 3$	7.41	1.71	11.63

Under the peaks in 0–5% at $0.8 < p < 1\,\mathrm{GeV}/c$ the random fraction is 4.0% (π), 8.3% (K) and 3.3% (p) with equal chance, and 1.9%, 2.5% and 0.7% with the earliest-time rule (Figs. 21a, 22a).

- Equal chance roughly doubles the random matches. It describes the HI region above $p \approx 0.8\,\mathrm{GeV}/c$, but it exceeds the SE yield in the valleys next to the peaks by up to a factor 3.08, which is impossible. The track's own hit must therefore win more often than an equal-chance draw gives.
- Earliest time wins never exceeds the SE yield (largest ratio 0.83). By construction, though, a random hit wins only when it is earlier than the own hit, so it fills the low-m^2 side and leaves the HI excess (SE/ME = 11.63).
- Production selects on ToT, not on time, so the two emulations bracket the in-time random matches that come from hits with a time: 3.3–7.3% of the TOF matches in 0–5%.

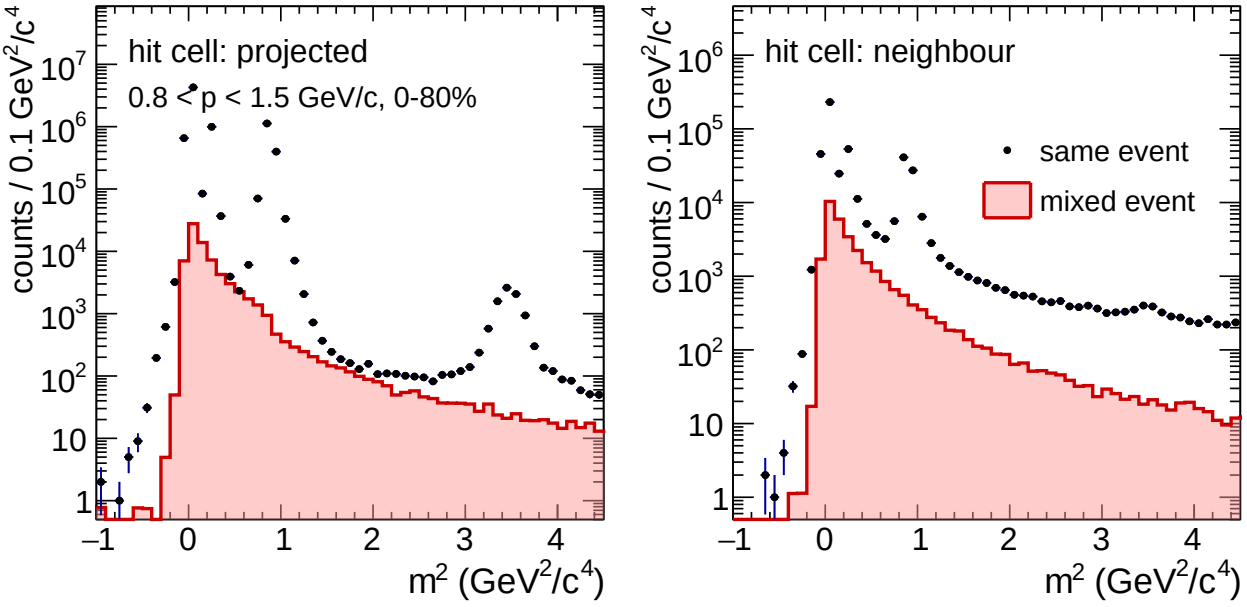
7.6.3 A late component flat in time

In time units the HI excess is a nearly flat plateau. Between 16 and 26 ns after the pion arrival time the SE rate falls only slowly, while the ME falls by about a factor 4. Only 0.45% of the hits that have a time in the picoDst arrive later than 25 ns. The plateau hits must therefore be among the hits without a time, i.e. hits that production did not match to any track in their own event. Such a hit enters the SE when production matches it to the track under study, because the track's BTofPidTraits then carries its time. In a partner event it has no time and cannot give an m^2, so no picoDst-based mixed event can contain it.

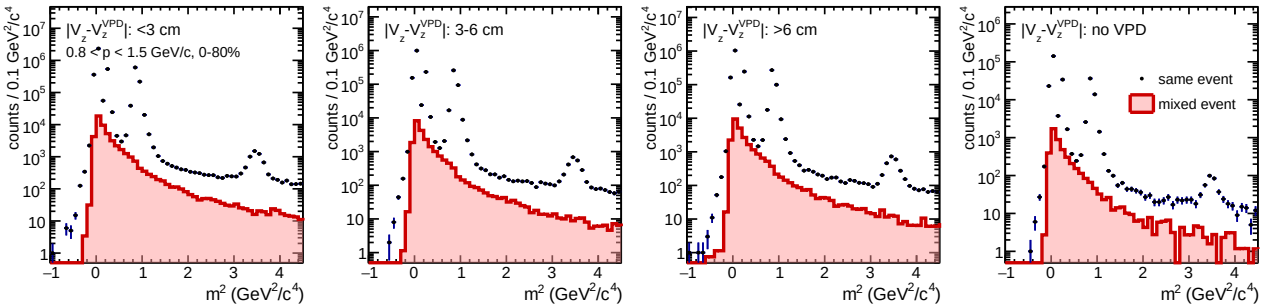
The plateau rate per projected track (Fig. 20b) rises only from 25 to 89 hits per ns per 10^6 tracks (factor 3.5) from 70–80% to 0–5% for $0.6 < p < 1.5\,\mathrm{GeV}/c$, and from 166 to 349 (factor 2.1) for $0.2 < p < 0.6\,\mathrm{GeV}/c$. The BTOF hit multiplicity, which follows refMult (Fig. 2a), grows by more than an order of magnitude over the same range (read from the refMult range of the classes, not computed per class). Most of the plateau is therefore unrelated to the triggered collision: electronic noise or hits from other bunch crossings in the TOF readout window.

Caveat

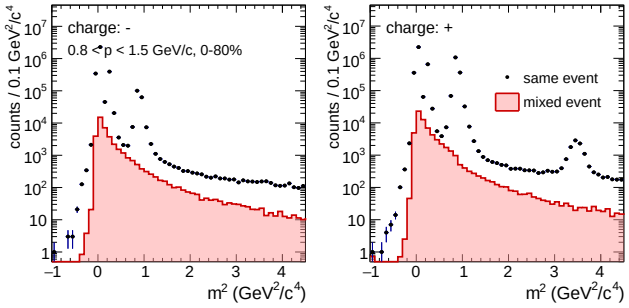
The picoDst cannot provide a complete mixed-event description of random matches: (1) it has no ToT, so the production choice among several hits can only be bracketed; (2) 65% of the BTOF hits have no time, among them the flat late component. Both are available in the MuDst (`StMuBTofHit`), where every hit has a time and a ToT; a MuDst test is being set up (Sec. 10).



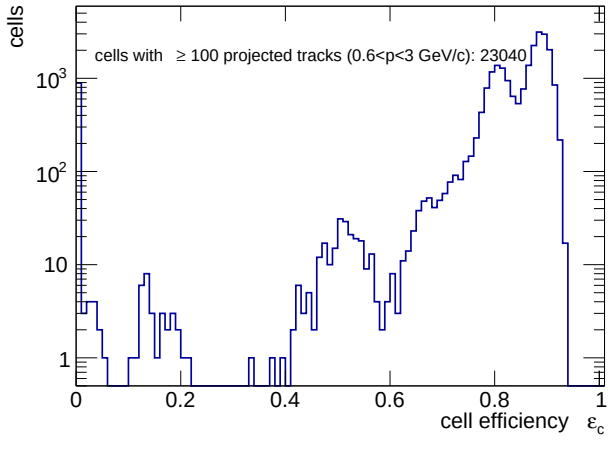
(a) SE (points) and overlay ME (red) m^2, $0.8 < p < 1.5\,\mathrm{GeV}/c$, 0–80%, for hits in the projected cell (left) and in a neighbour cell (right). Per 10^4 projected tracks is not defined for hit categories; ME is normalized per projected track in total.



(b) Same, split by $|V_z - V_z^{\mathrm{VPD}}|$: < 3, 3–6, $> 6\,\mathrm{cm}$ and no VPD vertex. ME normalized per projected track of each class.

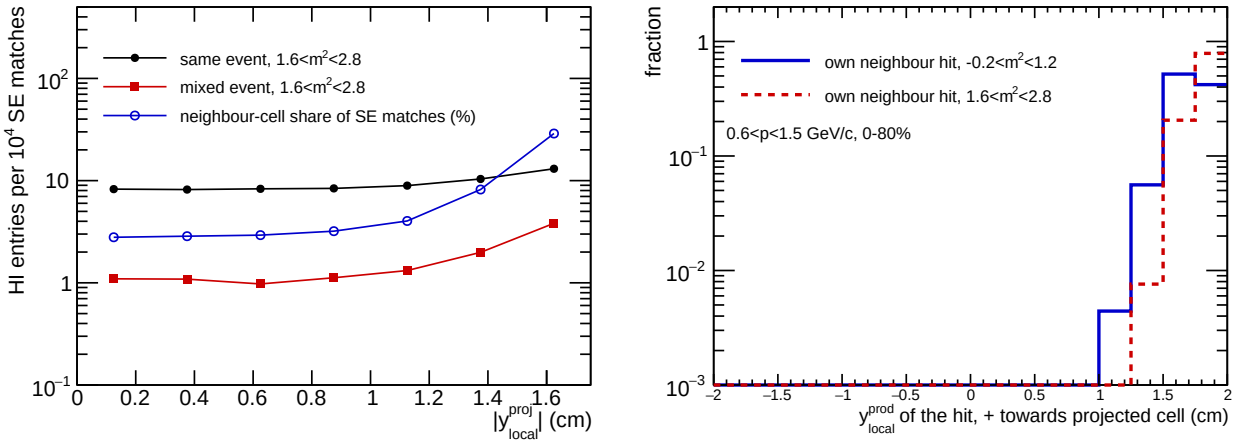


(c) Same, split by charge.

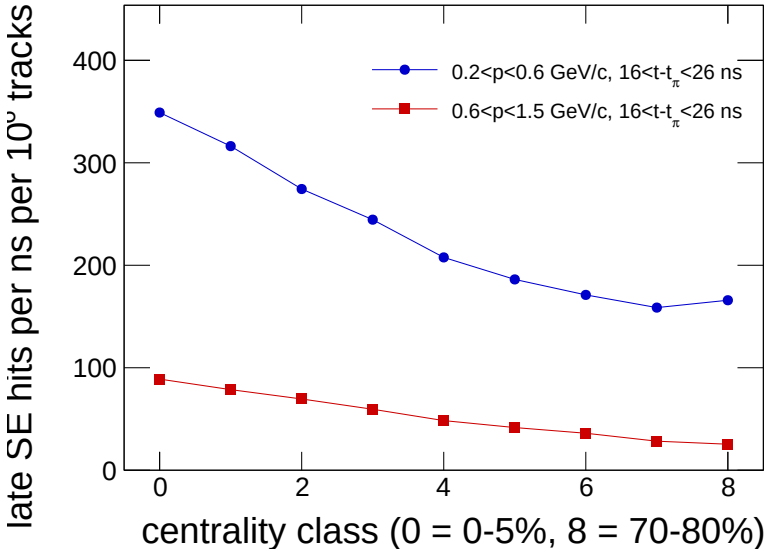


(d) Distribution of the cell efficiency ε_c ($0.6 < p < 3\,\mathrm{GeV}/c$), log scale.

Figure 19: Background diagnostics (v4, v5).

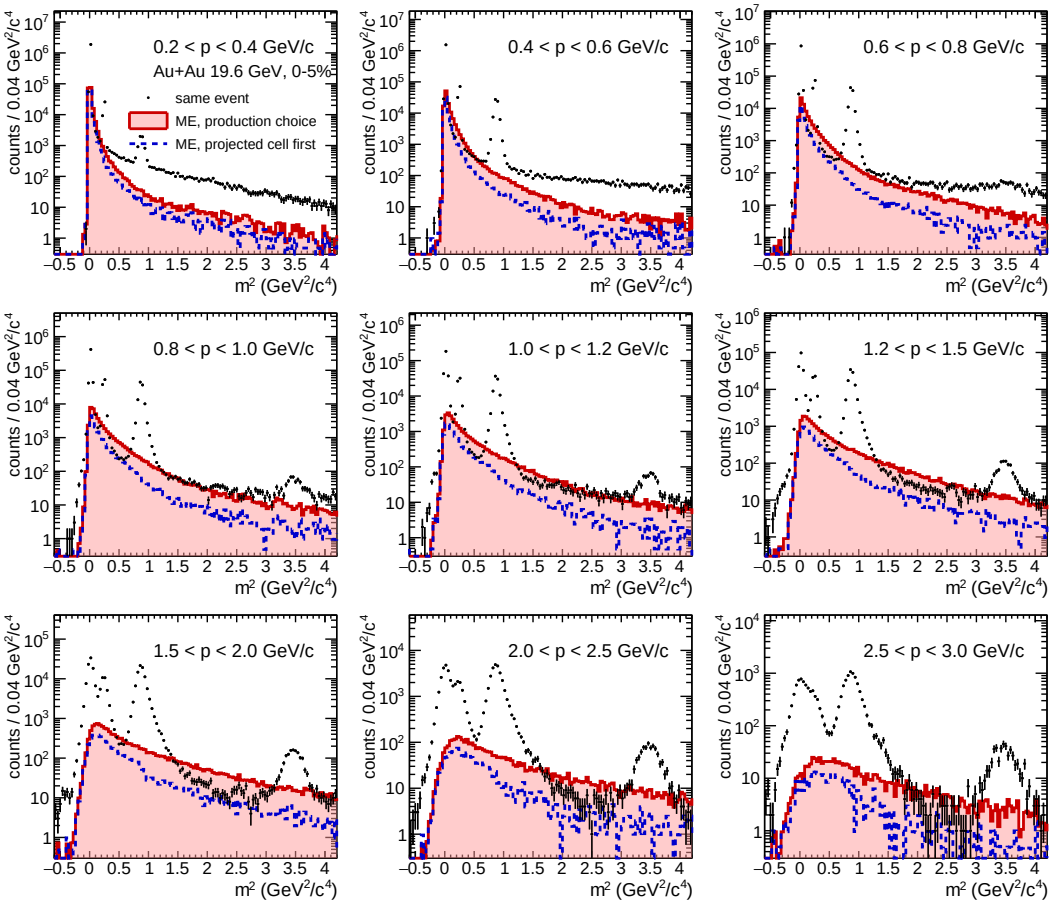


(a) Left: HI entries per 10^4 SE matches vs $|y_{\mathrm{local}}|$ of the projection for SE (black) and overlay ME (red), and the share of neighbour-cell matches (open blue, %). Right: production y_{local} of own neighbour hits (read by eye; the quantity is not used in the conclusions).



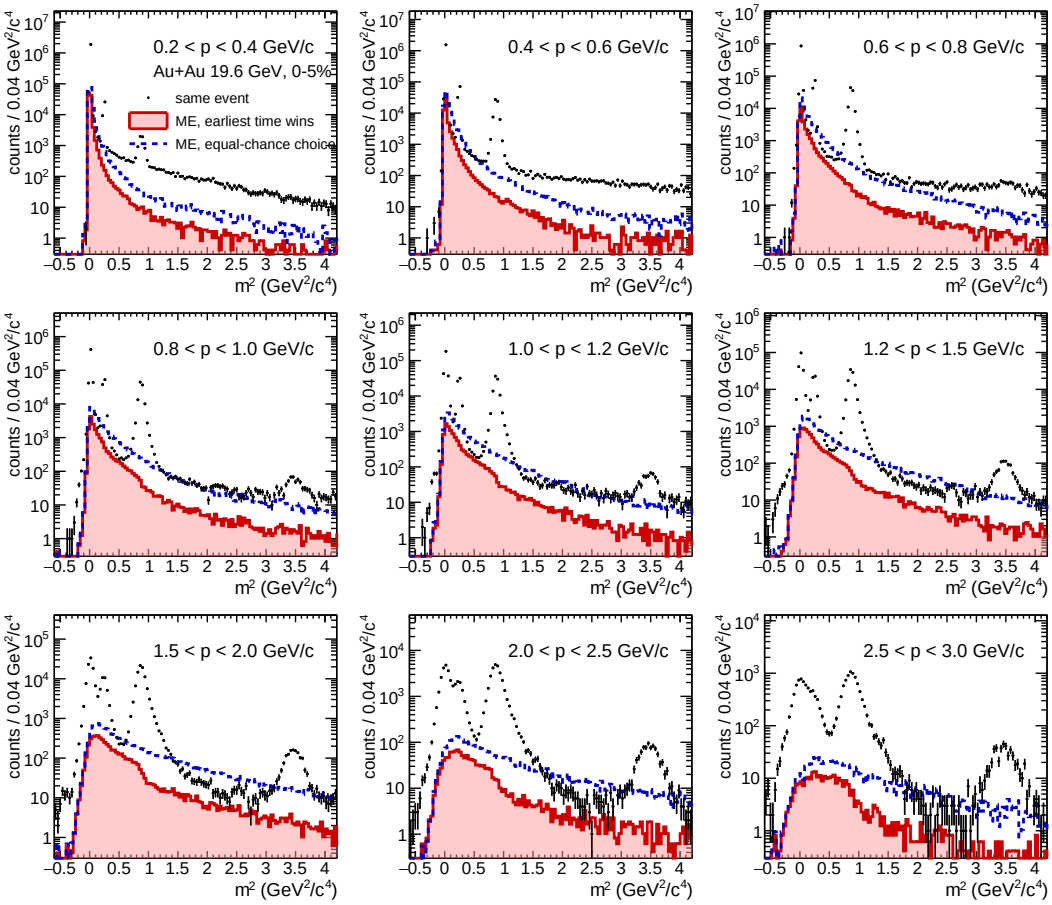
(b) SE hits arriving 16–26 ns after the pion time, per ns and per 10^6 projected tracks, vs centrality class, for $0.2 < p < 0.6$ (blue) and $0.6 < p < 1.5\,\mathrm{GeV}/c$ (red).

Figure 20: Neighbour-cell matches (v6) and the late plateau (v8).



(a) m^2 of projected QA tracks, 0–5%, nine momentum bins (log scale). Black: same event. Red filled: overlay ME with the equal-chance production choice, all hits, normalized per projected track. Blue dashed: overlay ME with the projected cell first (Sec. 7.5).

Figure 21: SE vs ME m^2, 0–5%, equal chance.



(a) m^2 of projected QA tracks, 0–5%, nine momentum bins (log scale). Black: same event. Red filled: overlay ME with the earliest time winning, timed partner hits only, normalized per projected track. Blue dashed: equal-chance choice.

Figure 22: SE vs ME m^2, 0–5%, earliest time wins.

7.7 Mixed event on MuDst

The picoDst stores a time only for BTOF hits that production matched to a track (35.3% of the hits, Sec. 7.6). The unmatched hits therefore cannot compete in a picoDst mixed event, and the late same-event background could not be reproduced (best picoDst result: SE/ME = 9.25 between the p and d peaks). The MuDst keeps every `StMuBTofHit` with its leading and trailing edge, so this section repeats the mixed event on MuDst. Two methods are used: an emulation of the production matching in our own maker (Sec. 7.7.3), and a re-run of the production makers themselves on the overlaid events (Sec. 7.7.4).

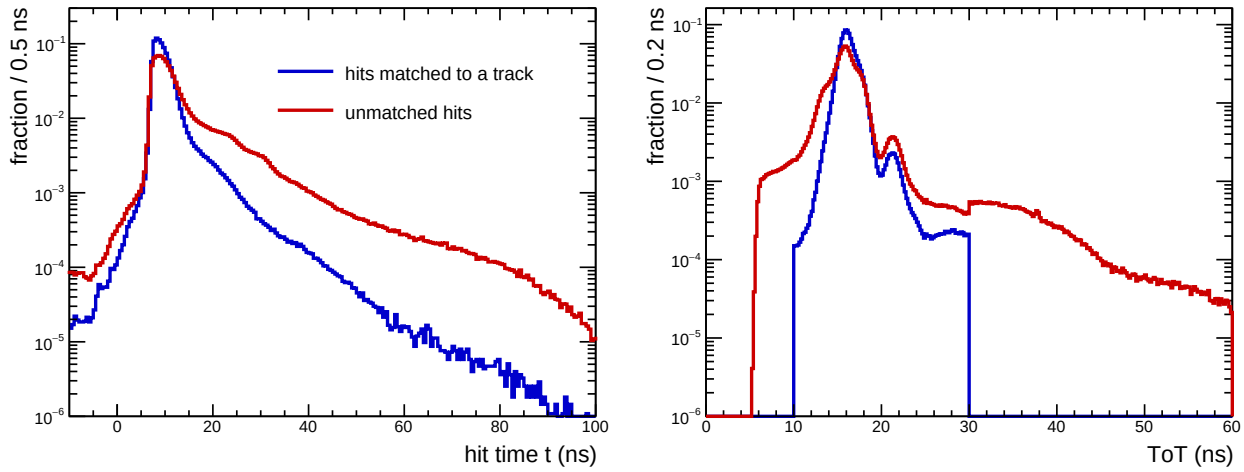
Sample. The P24iy MuDst are on HPSS only. Six MuDst files of three runs were restored through the Data Carousel (`scripts/mudst_restore.sh`, list `filelists/mudst/restore_test.txt`); they are the raw files of the picoDst sample. The two files of run 20088015 are not used: reading them floods the log with trigger-data decoding errors and the jobs stall. The four files used are `st_physics_20060065_raw_1000011`, `_5500010`, `st_physics_20073019_raw_4500013` and `_7000013`: 68459 events, of which 21160 pass the event selection of Sec. 7.5 and have a VPD start time. The m^2 numbers in this section use these events only; they are a test sample of two runs. Statistical uncertainties only.

7.7.1 Times of matched and unmatched hits

`macros/probeMuBTof.C` reads all hits of the four files (23653 selected events) in two passes. Pass 1 measures a per-cell offset as the median of $t_{\mathrm{LE}} - t_{\mathrm{tof}} - T_0$ over hits matched by production, where t_{tof} is the production time of flight and T_0 the event median of $t_{\mathrm{LE}} - t_{\mathrm{tof}}$. Pass 2 gives every hit the time $t = t_{\mathrm{LE}} - \mathrm{off}_{\mathrm{cell}} - T_0$. On matched hits t reproduces the production time of flight with an RMS of 0.54 ns (97% within 1 ns); 0.2% of the unmatched hits are in cells without an offset and are dropped. Of the hits, 2761120 are matched and 3929867 (59%) are unmatched.

Unmatched hits are systematically later (Fig. 23a). The fraction at 8–10 ns is 42% for matched and 27% for unmatched hits; at 15–26 ns it is 5.5% and 15.9%, and above 26 ns 1.3% and 9.5%. Per ns, unmatched hits are 7 times more frequent than matched hits at 30–60 ns. The late fraction of unmatched hits falls only mildly with the BTOF hit multiplicity (12% for the lowest, 8% for the highest classes). Their ToT distribution has the same mean as for matched hits (16.6 vs 16.4 ns). This is the behaviour of slow particles that are not reconstructed in the TPC, not of electronic noise; the hits are in the same bunch crossing by construction of the production.

All matched hits have $10 < \mathrm{ToT} < 30$ ns (0.00% outside, Fig. 23a right), while 3% of the unmatched hits are below 10 ns and 3.1% above 30 ns. Hits outside this window cannot end up as a calibrated match, so they never contribute to random matches.



(a) BTOF hits of the four MuDst files, normalized to unit area. Left: calibrated hit time t (0.5 ns bins, log scale). Right: ToT (0.2 ns bins). Blue: hits matched to a track by production; red: unmatched hits. The legend of the left panel applies to both. Unmatched hits have a long late tail: 9.5% above 26 ns against 1.3% for matched hits.

Figure 23: Time and ToT of BTOF hits in MuDst (`scripts/make_probe.C`).

7.7.2 Closure test: production matching and calibration re-run on MuDst

The SL24y makers `StBTofMatchMaker`, `StVpdCalibMaker` and `StBTofCalibMaker` can run on MuDst (`setMuDstIn()`). The chain (`macros/runTofClosure.C`) is `StMuDstMaker`, `St_db_Maker` (current database), the three makers with the y2019 geometry, and `StRoot/StTofClosureMaker`. That maker stores the production matching of every global track before the re-run and compares it afterwards. On 300 events of `st_physics_20060065_raw_1000011`:

- all 16451 production matches are found again in the same cell; there are 0 matches in another cell, 0 only in production and 0 only in the re-run;

- the re-run time of flight differs from production by a median of -14 ps (68% between -28 and 1 ps, 98% within ±133 ps); 88% of the tracks agree in $1/\beta$ within 0.005;
- the event start time differs by a median of 4 ps, and 51% of the events agree within 5 ps.

The residual differences are small compared with the BTOF resolution; they are attributed to calibration tables updated after the production (not verified: the database time stamp was not pinned to the production).

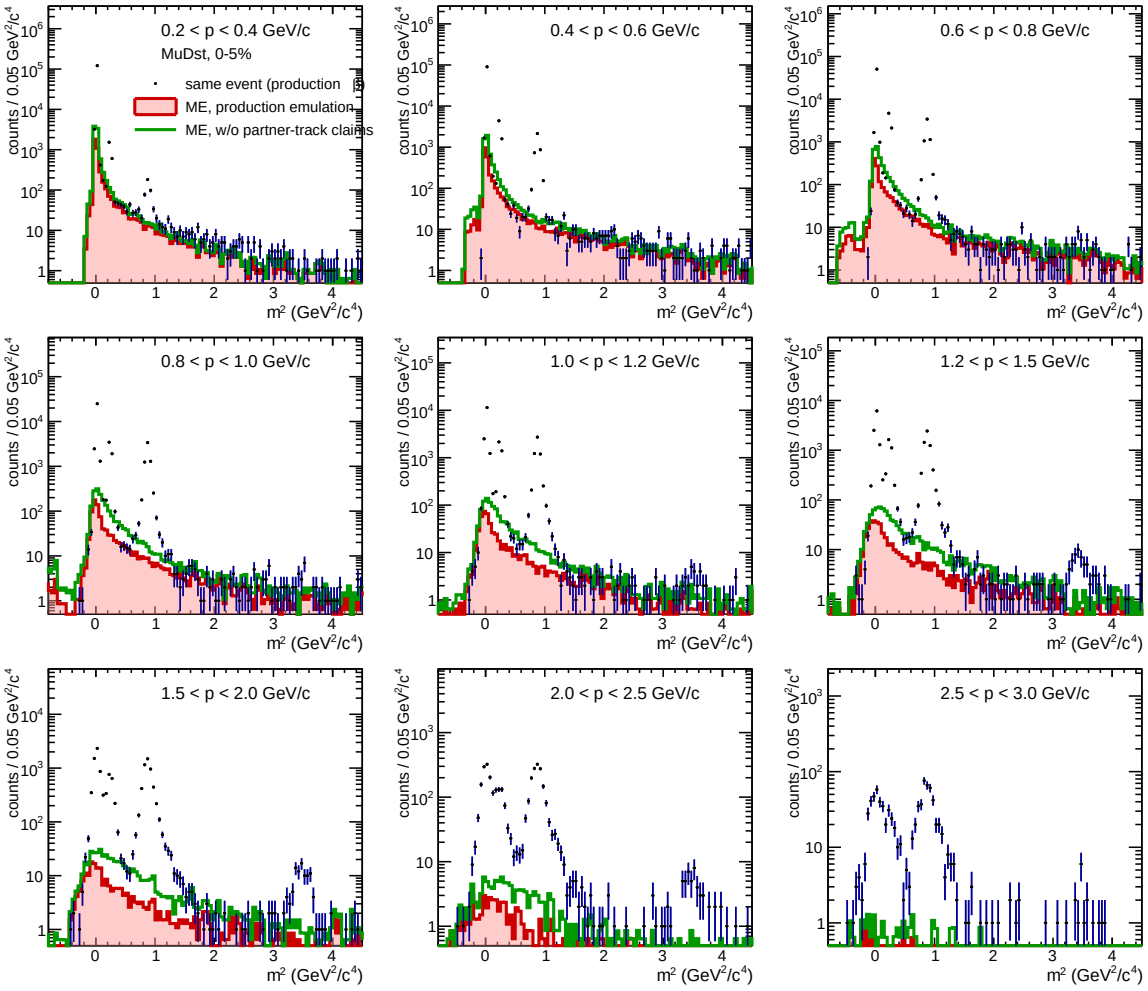
Choice

The re-run of the production BTOF makers on MuDst reproduces the production matching exactly and the time of flight to within tens of ps. It is therefore used for the mixed event in Sec. 7.7.4, so that same event and mixed event share the same matching and calibration.

7.7.3 Method 1: emulation of the production matching

`StRoot/StTofMEMuMaker` (derived from `StTofMEMaker`, same projection and mixing classes) emulates the SL24y matching rule on the overlaid hit map. Every production-valid global track (flag, $n_{\mathrm{HitsFit}} \geq 15$, ratio ≥ 0.52, outer helix) is projected; candidates are the projected cell and its two neighbours in the module. A cell claimed by more than one track is discarded, and among several candidates the largest ToT (below 40 ns) wins. Hit times are our own calibration (t_{LE} minus per-cell offset minus event T_0, RMS 0.43 ns against production). On the same event, the emulation picks the production hit for 98.3% of the production matches.

In the mixed event, the hits of a partner event are overlaid on the hit map of the event, and hits of the two events in the same cell are merged (earlier leading edge, larger ToT). The partner's own production-valid tracks also claim the cells of the partner hits. A partner hit made by a reconstructed particle is thus discarded as shared, exactly as in its own event. Without these claims the random fraction is 6.0% in 0–5% (green line in Fig. 24a). With them it is 2.7% (0–5%) and 1.6% (0–80%). The same-event histogram uses the production β.



(a) m^2 of QA tracks, 0–5%, nine momentum bins, four MuDst files (log scale). Black points: same event (production β). Red filled: mixed event with the production emulation including the partner-track claims, normalized per QA track. Green line: the same without partner-track claims. Between the p and d peaks SE/ME = 1.29 (0.2–3 GeV/c).

Figure 24: MuDst mixed event, emulation of the production matching, 0–5% (`scripts/make_mu.C`).

Caveat

The emulation uses our own hit times (RMS 0.43 ns, no slewing or z_{local} correction). Random matches to pion-like partner hits are smeared about three times wider in m^2 than in the same event. In the thin bins just below and beside the pion and kaon peaks the mixed event exceeds the same event bin by bin, up to ME/SE = 3.70 in 0–5% ($0.4 < p < 1.5$ GeV/c). This is the motivation for method 2.

7.7.4 Method 2: production makers re-run on overlaid events

`StRoot/StTofMEOverlayMaker` (macro `macros/runTofMEOverlay.C`, one HTCondor job per file) runs after the chain of Sec. 7.7.2, which first re-runs the three production makers on the event itself; this gives the same event. Mixing classes are 14 V_z bins × 9 refMult classes per run, with the last 5 events of a class as partners. For each partner:

1. the partner hits are appended to the `StMuBTofHit` array of the event, with leading and trailing

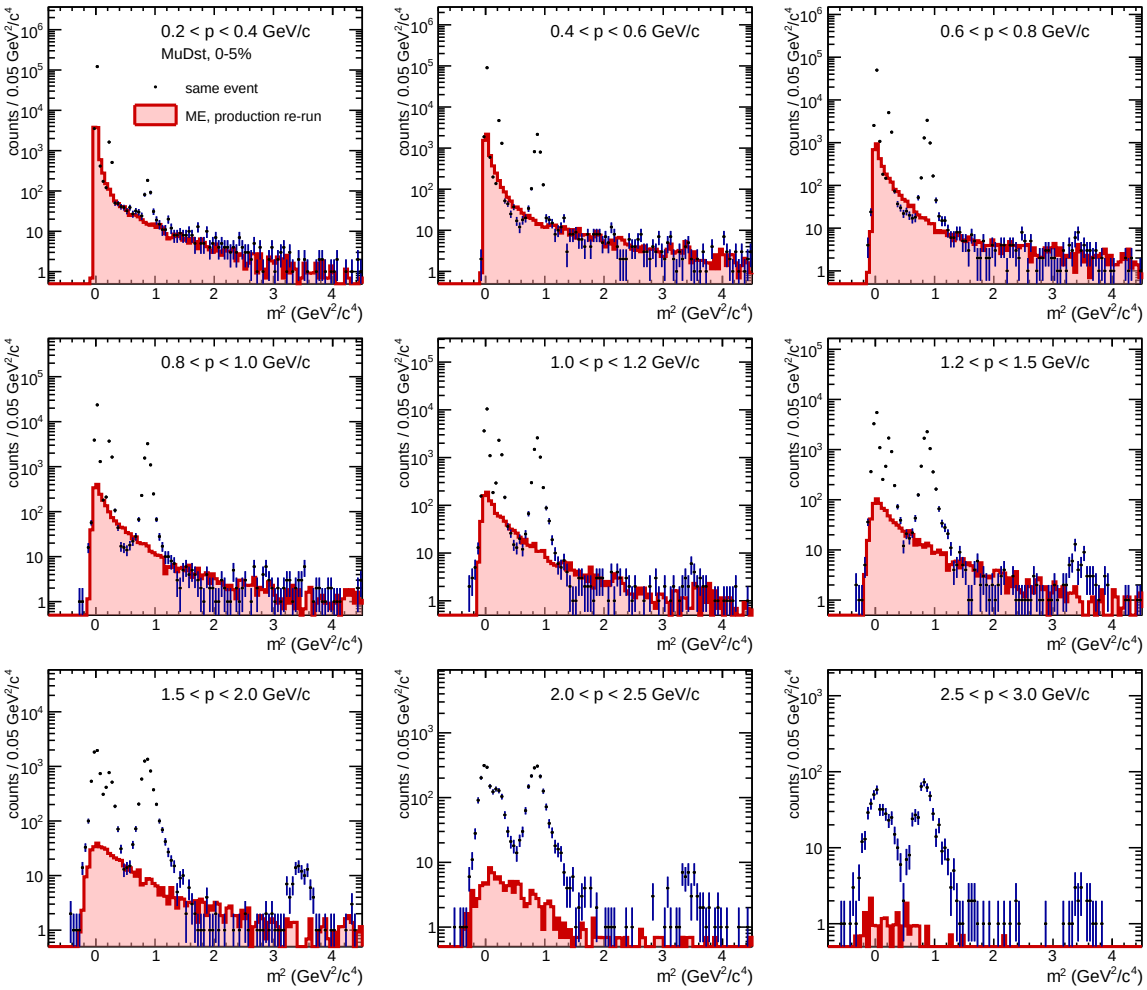
edges shifted by $t_{\mathrm{start}}^{\mathrm{event}} - t_{\mathrm{start}}^{\mathrm{partner}}$ (VPD start times from `StBTofHeader`), so that each partner hit keeps its time of flight. A partner hit in a cell that already has an overlapping hit ($|\Delta t_{\mathrm{LE}}|$ below the ToT of the earlier hit) is merged into it: earlier leading edge, later trailing edge;

2. `StBTofMatchMaker`, `StVpdCalibMaker` and `StBTofCalibMaker` are run again on the overlaid event;

3. a QA track matched to an appended hit, or to a merged hit whose leading edge came from the partner, is a random match. Its m^2 is computed from the re-run time of flight and path length, exactly as for the same event;

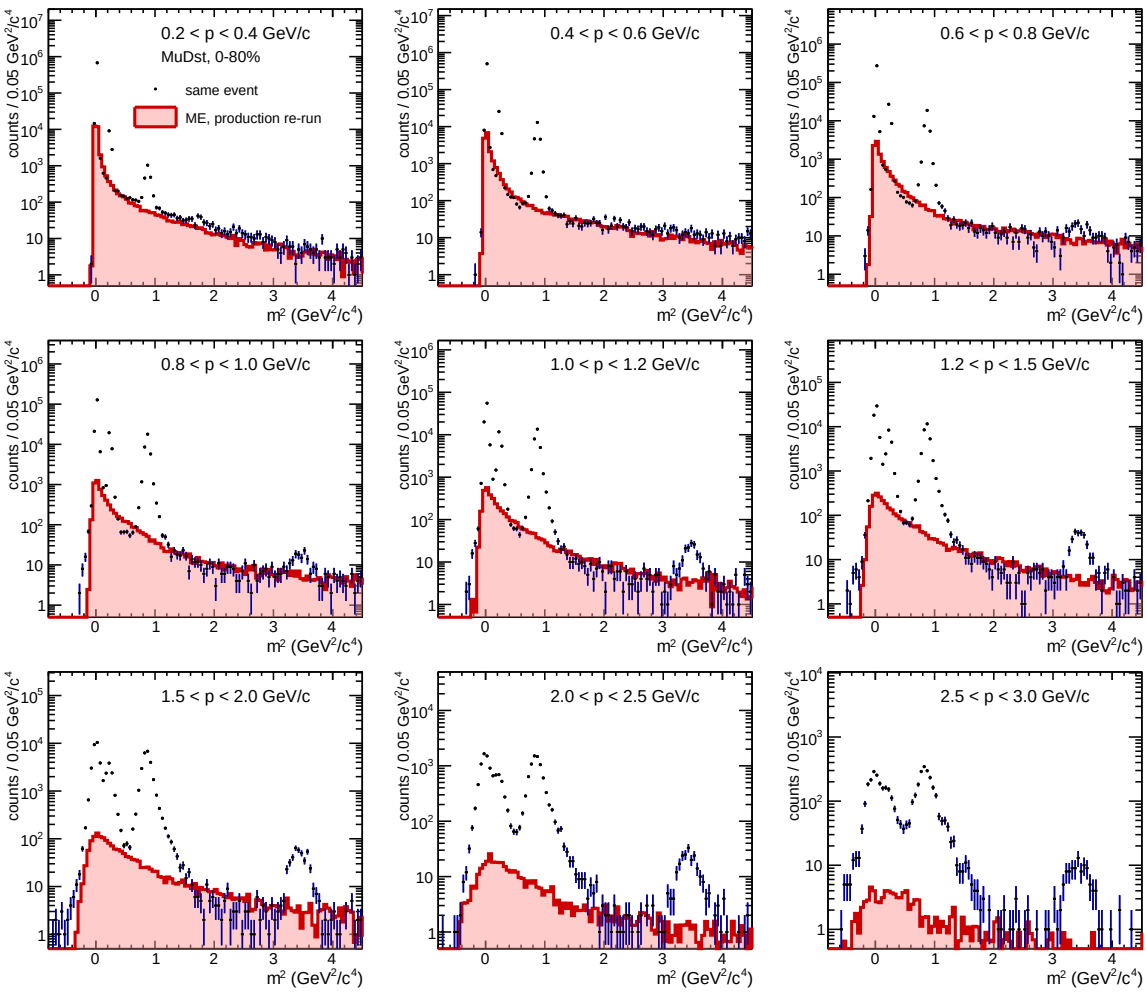
4. the hit array of the event is restored.

The mixed event is normalized per QA track in each (centrality, p) bin. In the overlays 56.9% of the QA tracks have a match, compared with 56.4% in the same event; 2.7% of the same-event matches are replaced by a partner hit or lost.



(a) m^2 of QA tracks, 0–5%, nine momentum bins, four MuDst files (log scale). Black points: same event. Red filled: mixed event with the production makers re-run on the overlaid hits, normalized per QA track. Both use the re-run calibration. SE/ME between the p and d peaks is 0.92 (0.2–3 GeV/c); the largest bin-by-bin ME/SE at $0.4 < p < 1.5$ GeV/c is 2.14.

Figure 25: MuDst mixed event, production makers re-run on overlaid events, 0–5% (`scripts/make_ov.C`).



(a) As Fig. 25a for 0–80% centrality.

Figure 26: MuDst mixed event, production makers re-run, 0–80%.

With the common calibration the spike of method 1 beside the pion peak disappears, and the mixed event follows the shape of the same-event background at all m^2, including the region between the p and d peaks and the valleys (read by eye). Its level, however, is too high: the random fraction is 6.0% (0–5%) and 3.6% (0–80%), equal to method 1 without partner-track claims, and at 0.6–2 GeV/c the mixed event is above the same event between the p and d peaks (SE/ME below 1 in every bin, Table 11).

Caveat

In method 2 only the partner hits are added, not the partner tracks. A partner hit made by a reconstructed particle is therefore never discarded as shared, which it would be in a real event, where its own track also claims it. Method 2 overestimates the random matches by this amount. Adding the partner tracks to the overlay would also make them compete for the hits of the event (a denser event); this is not yet done.

7.7.5 Comparison and current result

Table 11: MuDst random matches per momentum bin. "Emul. = production emulation with partner-track claims (Sec. 7.7.3); "Overlay = production makers re-run on the overlaid hits, partner tracks absent (Sec. 7.7.4). f = ME/SE of all TOF matches (%); HI = SE/ME for $1.6 < m^2 < 2.8\,\mathrm{GeV}^2/c^4$; N_{HI} = SE entries in HI; π/K/p = ME/SE under the peaks (%).

							emul.			overlay		
cent.	p (GeV/c)	f_{emul}	$\mathrm{HI}_{\mathrm{emul}}$	N_{HI}	f_{ov}	$\mathrm{HI}_{\mathrm{ov}}$	π	K	p	π	K	p
0-5%	0.2–0.4	3.2	2.01	118	7.3	1.44	2.6	11.9	21.3	6.5	17.2	27.4
0-5%	0.4–0.6	2.7	1.21	129	5.9	1.00	1.9	4.7	2.3	4.7	9.8	3.1
0-5%	0.6–0.8	2.5	0.98	76	5.3	0.84	1.7	3.0	1.3	3.9	7.5	1.9
0-5%	0.8–1	2.4	1.07	54	5.2	0.78	1.7	2.1	0.9	3.6	6.6	1.9
0-5%	1–1.2	2.3	1.03	43	5.0	0.76	1.6	1.8	0.7	3.4	5.9	1.6
0-5%	1.2–1.5	2.2	1.19	46	4.9	0.67	1.4	1.6	0.7	3.1	5.6	1.5
0-5%	1.5–2	2.0	1.10	25	4.6	0.47	1.2	1.3	0.4	2.7	5.3	1.3
0-5%	2–2.5	1.9	2.61	19	4.3	1.31	0.9	1.9	0.3	2.3	4.7	1.4
0-5%	2.5–3	1.9	5.56	13	4.3	2.42	1.0	0.9	0.1	2.4	3.7	1.3
0-80%	0.2–0.4	1.9	2.16	474	4.3	1.63	1.5	7.7	13.9	3.8	11.1	17.6
0-80%	0.4–0.6	1.6	1.42	533	3.4	1.16	1.1	2.9	1.5	2.7	6.0	1.9
0-80%	0.6–0.8	1.5	1.10	292	3.2	0.86	1.0	1.8	0.8	2.3	4.8	1.2
0-80%	0.8–1	1.4	1.13	186	3.1	0.82	1.0	1.3	0.6	2.1	4.0	1.1
0-80%	1–1.2	1.4	1.05	133	3.0	0.71	0.9	1.1	0.4	1.9	3.8	1.0
0-80%	1.2–1.5	1.4	1.23	146	3.0	0.68	0.8	1.0	0.4	1.8	3.5	0.9
0-80%	1.5–2	1.3	1.41	106	2.9	0.65	0.7	0.9	0.3	1.7	3.2	0.8
0-80%	2–2.5	1.1	3.17	68	2.6	1.40	0.5	1.0	0.2	1.5	3.2	0.8
0-80%	2.5–3	1.2	5.52	34	2.7	1.99	0.5	0.6	0.2	1.5	3.0	0.7

Decision needed

Current result (option chosen for this draft): method 1 with partner-track claims is the central value, 2.7% of the TOF matches in 0–5% and 1.6% in 0–80% ($0.2 < p < 3\,\mathrm{GeV}/c$). Method 2 is an upper bound, 6.0% and 3.6%. The difference is a systematic uncertainty until the partner tracks are included in method 2. The fractions under the PID peaks per momentum bin are listed for both methods in Table 11.

Both methods describe the same-event background between the p and d peaks far better than any picoDst mixed event: SE/ME there is 1.29 (method 1) and 0.92 (method 2) in 0–5%, against 9.25 on picoDst. The late background is therefore made of random matches to real, mostly unreconstructed slow particles. It is missing in the picoDst mixed event because these hits have no time there. Above $2\,\mathrm{GeV}/c$ and below $0.4\,\mathrm{GeV}/c$ the HI window has few same-event entries per bin (N_{HI} in Table 11), so these bins need the larger sample (about 100 MuDst files, restore requested).

8 Cuts adopted and inputs for the mixed-event step

What this QA implies for the mixed-event matching:

- With 2.94 stored hits per TOF-matched QA track and a matching window of one pad in y and $\pm 3.05\,\mathrm{cm}$ in z, the random-match probability will depend strongly on event multiplicity; events must be mixed within multiplicity classes (refMult or number of BTOF hits) and V_z classes.
- The picoDst stores only the cell ID of each BTOF hit, so matching is done at the cell level with the projection of Sec. 7 (implemented).

Table 12: Selection used in this QA (defaults in `macros/runTofMEQA.C` and the constructor of `StTofMEQAMaker`).

Level	Cut	Status
Event	trigger $\in$ {640001, 640011, 640021, 640031, 640041, 640051}	applied
Event	$\lvert V_z \rvert < 70\,\mathrm{cm}$	applied
Event	$V_r < 2\,\mathrm{cm}$	applied
Event	$\lvert V_z - V_z^{\mathrm{VPD}} \rvert$	not applied
Event	pile-up / bad-run rejection	not implemented
Track	primary, $p_T > 0.15\,\mathrm{GeV}/c$	applied
Track	nHitsFit ≥ 15, nHitsFit/nHitsMax > 0.52	applied
Track	gDCA $< 3\,\mathrm{cm}$	applied
Track	$\lvert \eta \rvert < 1$	applied
TOF	btofMatchFlag > 0	applied
TOF	$\beta > 0$	applied (TOF-matched sample)
TOF	$\lvert y_{\mathrm{local}} \rvert$, $\lvert z_{\mathrm{local}} \rvert$	none beyond production ($\lvert z_{\mathrm{local}} \rvert < 3.05\,\mathrm{cm}$)

- Dead trays/modules are identical between mixed events because mixing is restricted to one run (implemented).
- The y_{local} side-bands ($\lvert y_{\mathrm{local}} \rvert > 2.5\,\mathrm{cm}$) provide an independent same-event estimate of the random-match level that the mixed-event result can be compared to.

Decision needed

(1) Is the minimum-bias trigger list correct for P24iy, or should the analysis use 640051 only? (2) Should a $\lvert V_z - V_z^{\mathrm{VPD}} \rvert$ cut be used, given the VPD vertex is available for only 92.9% of events? (3) Mixing classes: refMult (used now) or number of BTOF hits, and is a 10 cm V_z class adequate? (4) Which matching rule should the final random-match correction use: production-like cell$\pm$1 (default proposal) or cell only? (5) Low-p_T projection: accept the DCA-helix projection, or add an energy-loss correction / use the production cell for same-event matches?

9 Audit

9.1 Checks that passed

- **MuDst production re-run:** 16451 of 16451 production matches reproduced in the same cell, 0/0/0 different/lost/new; time of flight median -14 ps (`StTofClosureMaker`, 300 events, `scripts/make_probe.C`, `logs/probe_numbers.txt`).
- **MuDst emulation:** the emulated production choice picks the production hit for 98.3% of the production matches (`scripts/make_mu.C`, 21116 events).
- **Overlay bookkeeping:** matched fraction of QA tracks 56.4% in the same event and 56.9% in the overlays; 2.7% of the same-event matches are replaced or lost (`scripts/make_ov.C`).
- **Code identity:** md5 sums of `StTofMEQAMaker.{h,cxx}`, `runTofMEQA.C` and `drawTofMEQA.C` on the node equal those of the local copies (`logs/provenance.txt`, `logs/md5_local.txt`). The library was rebuilt (`cons`) before the run that produced `output/qa_test_10k.root`.

- **Event count:** 10000 events processed out of 17041 in the file, 1 file read (run log, `logs/runinfo.txt`); the first bin of `hEvtCut` equals the entries of `hVzAll` (10 000; PASS).
- **Cut-flow consistency:** entries of `hVz` (selected events) equal the last applied cut-flow bin (5 501; PASS); track counts in Table 4 decrease monotonically (PASS) (`scripts/make_numbers.C`, full test sample).
- **Cell decoding:** no matched track points to a module without hits (0 entries), consistent with the same `btofCellId/id()` encoding for traits and hits.
- **PID scale:** fitted m^2 peaks of π, K, p agree with PDG within $0.005\,\mathrm{GeV}^2/c^4$ (`scripts/make_numbers.C`, $0.4 < p < 1.0\,\mathrm{GeV}/c$).
- z_{local} **window:** the edges at $\pm 3.05\,\mathrm{cm}$ match `mZLocalCut=3.05` in `StRoot/StBTofMatchMaker/StBTofMatchMaker.cxx` of SL24y (source inspection).
- **API:** picoDst getters and compression factors were read from the SL24y `StRoot/StPicoEvent` headers (source inspection).
- **Geometry fit:** mean fit residual 0.0029 cm over 3699 modules; rotated-copy closure 0.297 cm mean (`macros/makeGeom.C`, `logs/makeGeom_test.log`).
- **Production neighbour rule:** the cell$\pm$1 acceptance and `mWidthPad`=3.45 cm were read in `StBTofMatchMaker.cxx` (SL24y); the fitted pitch agrees.
- **Same input sample:** the mixed-event test selects the same 5 501 events as the QA. It has 960 080 QA tracks, 77 fewer than the QA maker (960 157), because the test build compared nHitsFit/nHitsMax with the double 0.52 instead of the float 0.52 used in the QA maker (tracks with ratio exactly 13/25 were dropped). Fixed before the 1M-event productions.
- **1M merge:** all 15 job logs end normally and contain the maker summary; the merged file has 1 011 835 events read (`scripts/merge_v3.sh`, `logs/hadd_v3.log`); the job file lists contain no duplicate file (checked on the node).
- **Path length and m^2:** for tracks whose production hit is chosen, L agrees with $\beta_{\mathrm{prod}} c\, t_{\mathrm{prod}}$ (mean 0.001 cm, RMS 0.26 cm) and m^2 with production (RMS $0.0007\,\mathrm{GeV}^2/c^4$) (`hDLVsP`, `hDm2VsP`, `scripts/make_m2.C`).
- **Production hit choice:** largest ToT among candidate cells, then closest y, read in `StBTofMatchMaker.cxx` (SL24y, lines 782–833).
- **Background diagnostics:** every v4–v8 job log ends normally; all passes read the same 1 011 835 events (v6: 25 000 per run) (`scripts/merge_v4.sh`–`merge_v8.sh`).
- **Overlay ME does not exceed SE:** largest ME/SE over m^2 bins is 1.07 in 0–5%, $0.4 < p < 1.5\,\mathrm{GeV}/c$, against 3.8–5.5 without overlay (`scripts/make_m2.C`). It reaches 1.21 at $0.2 < p < 0.4\,\mathrm{GeV}/c$ (Table 8).
- P_{ME} **= occupancy:** slope of P_{ME} vs N_{hits} is 0.99 times 1/23040 (`scripts/make_me.C`, test sample).

9.2 Not verified (assumptions)

- MuDst: the database time stamp of the re-run is not pinned to the production; the residual ps-level differences are attributed to later calibration updates (not checked).
- MuDst: the restore of the event hit array after each overlay is not tested directly; the overlay matched fraction (56.9% vs 56.4%) is consistent with it.
- MuDst: 21160 (overlay) and 21116 (emulation) events are used; the difference comes from the start-time requirement (VPD start vs own T_0), not checked event by event.
- MuDst: the statement that the overlay ME follows the SE shape and that the pion-side spike disappears is read by eye (Fig. 25a); `tables/m2_v9.tex` is a stored output of `scripts/make_m2.C`, not regenerated by `build.sh`.
- Minimum-bias trigger IDs other than 640051 (not present in the test file).
- That the picoDst keeps all BTOF hits of an event (only inferred from equal mean/RMS of hit count and `btofTrayMultiplicity`).
- Meaning of trigger IDs 6, 35, 36, 37 and the 6406xx/6408xx IDs.
- Representativeness of run 20093036 for the dataset; no run-by-run QA.
- Catalog file/event counts are copied from the catalog query output in `logs/catalog_P24iy.txt`.
- The availability of NFS replicas was checked only for the latest runs (first 3000 catalog entries ordered by run number), not for the whole list.
- The geometry is fitted and the ME test is run on the same events (test sample). For the 1M sample the refitted geometry agrees with it to 0.0005 cm (`scripts/make_me_runs.C`), so this is checked for the runs used.
- The time distribution of BTOF hits without a stored time (43.2% of the chosen partner hits have one) is the same as for timed hits (all-hits normalization of Sec. 7.5).
- StRefMultCorr centrality and pile-up definitions for Run19 19.6 GeV were used as provided (not re-derived).
- ToT of a hit is independent of its origin (equal-chance emulation, Sec. 7.6); earliest time as a ToT proxy.
- The growth of the BTOF hit multiplicity between 70–80% and 0–5% is taken from refMult, not computed per class.
- Shape statements on the m^2 overlays (position of the random peak, where ME follows SE) are read by eye.
- Mixed-event independence of random hits on the track (no correlation of noise or afterpulses with real hits).
- Visual statements (shape of y_{local} tails, ϕ dip near -0.6, η dip of the efficiency at $\eta \approx 0$) are read from the figures, not fitted.

10 Open issues

Ordered by importance for the next step.

decision Matching rule for the final correction (cell±1 vs cell) and mixing classes.

1. Improve the low-p_T projection (energy loss, or outer helix not available in picoDst).
2. Translate P_{ME} into a random-match correction of the TOF-matched yield and m^2 background (needs the time of the random hit, not stored in picoDst).

decision MuDst random-match result: central value (emulation with partner-track claims) vs the production re-run on overlaid hits, which lacks the partner tracks (Sec. 7.7). Next: add the partner tracks to the overlay.

3. MuDst statistics: restore of 98 files of 14 runs requested (`filelists/mudst/restore_100.txt`); one job per file. Run 20088015 needs a workaround for the trigger-data decoding errors.
4. Understand the slight p_T dependence of P_{ME} seen in the 1M sample.
5. Process the three removed files (030, 033, 040) if the full 57-file sample is needed.

decision Confirm the trigger list and decide on VPD-vertex and pile-up cuts.

6. Scale beyond 1M events if needed (one HTCondor job per run, about 70k events in 35–40 min; avoid the qgp017 server).
7. Understand the reduced TPC track density near $\phi \approx -0.6$, $\eta > 0$.

11 Glossary

BTOF Barrel Time-of-Flight: 120 trays × 32 MRPC modules × 6 pads (cells), $|\eta| \lesssim 0.9$.

VPD Vertex Position Detector; start time and V_z^{VPD}.

picoDst STAR compact analysis format (`StRoot/StPicoEvent`).

refMult / gRefMult Reference multiplicity from primary / global tracks in $|\eta| < 0.5$.

nBTOFMatch Number of primary tracks matched to BTOF (stored per event).

btofTrayMultiplicity Number of BTOF hits in the event (stored per event).

btofMatchFlag 0 no match, 1 one-to-one, 2 one hit shared by several track candidates.

y_{local}, z_{local} Track projection relative to the centre of the matched pad.

gDCA Distance of closest approach of the global track to the primary vertex.

$\varepsilon_{\mathrm{match}}$ TOF-matched ($\beta > 0$) QA tracks divided by all QA tracks.

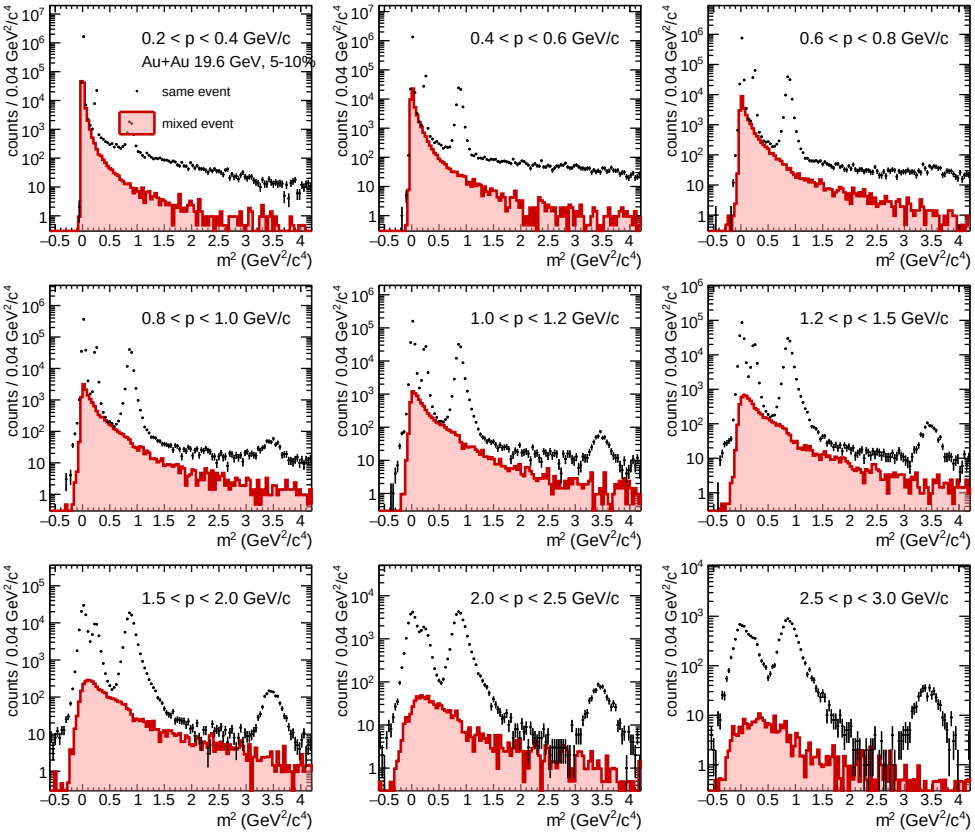
P_{SE}, P_{ME} Probability of a hit in the projected cell (or cell±1) in the same / a mixed event.

f_{rand} $P_{\mathrm{ME}}/P_{\mathrm{SE}}$, estimated random fraction of same-event matches.

Mixed event Pairing tracks of one event with BTOF hits of another event of similar class.

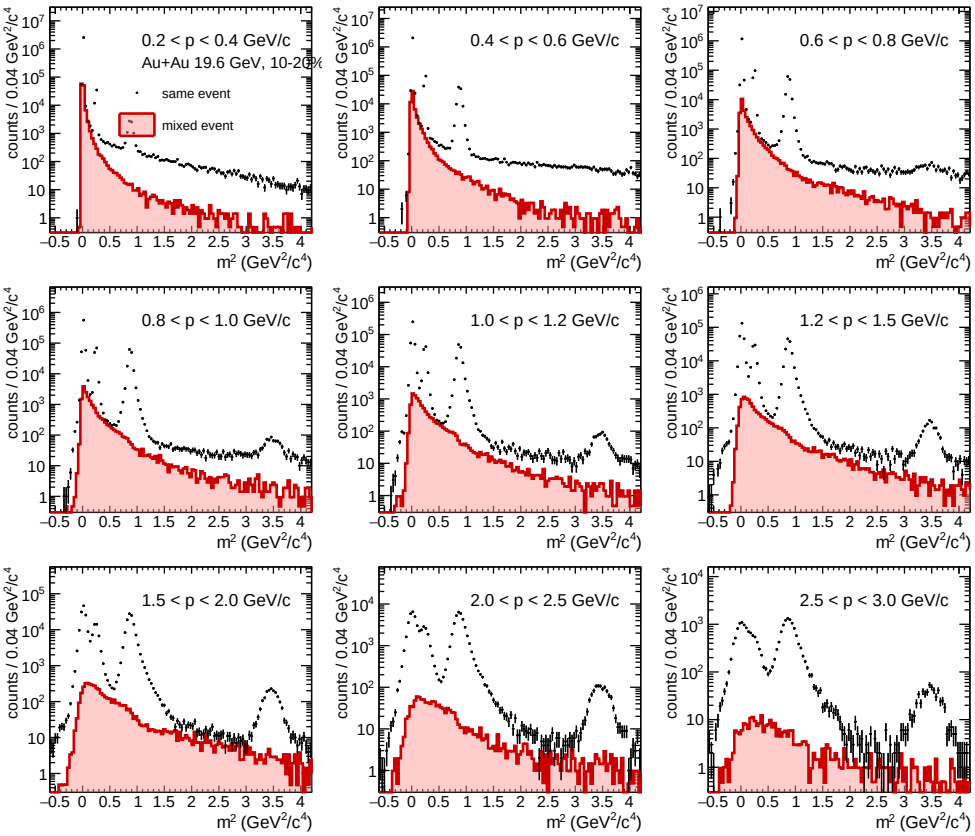
A SE vs ME m^2 for the other centrality classes

Same content as Figs. 15a–17a (`scripts/make_m2.C`, 1M sample).



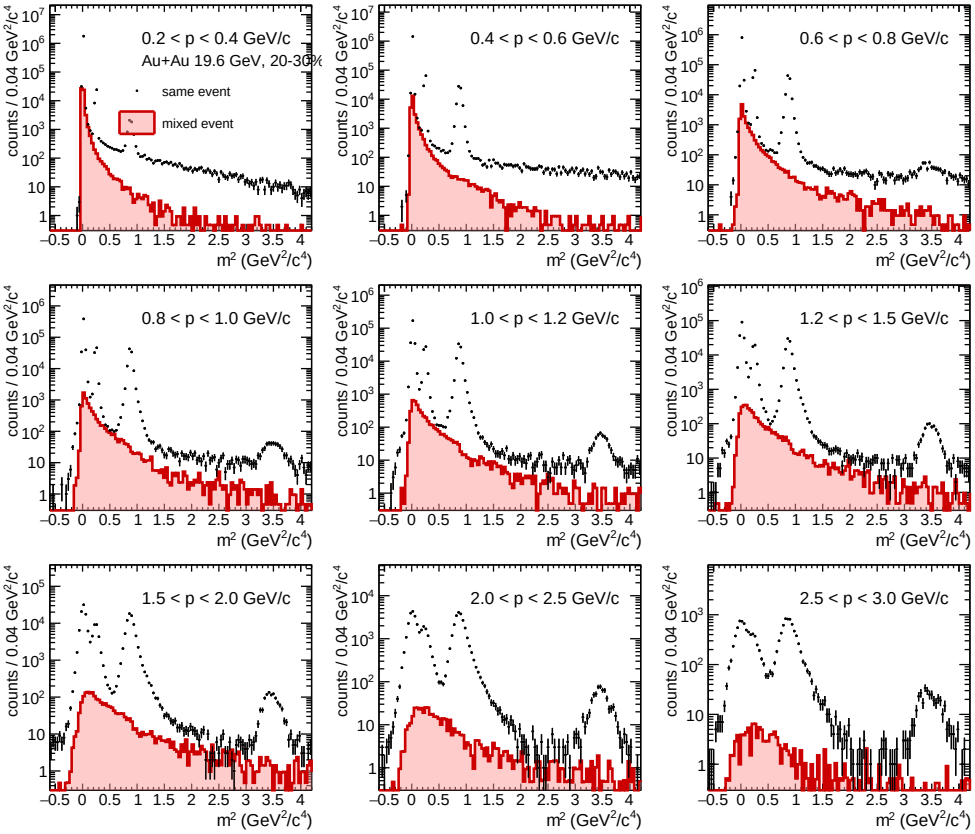
(a) m^2 of projected QA tracks, 5–10%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 27: SE vs ME m^2, 5–10%.



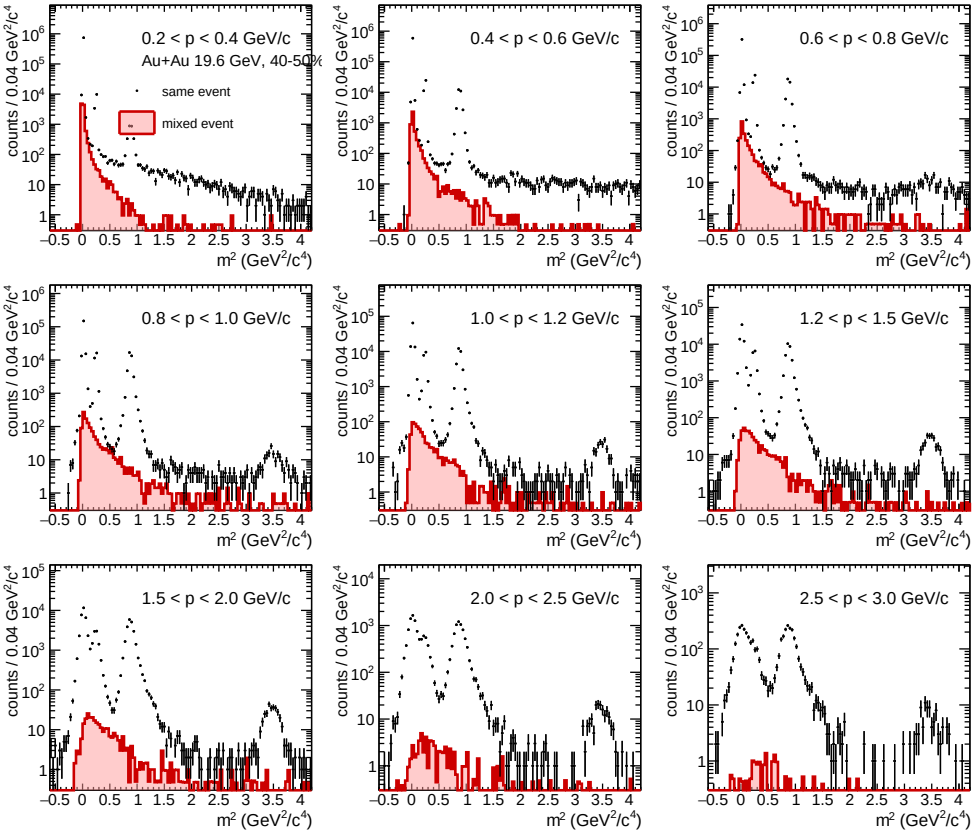
(a) m^2 of projected QA tracks, 10–20%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 28: SE vs ME m^2, 10–20%.



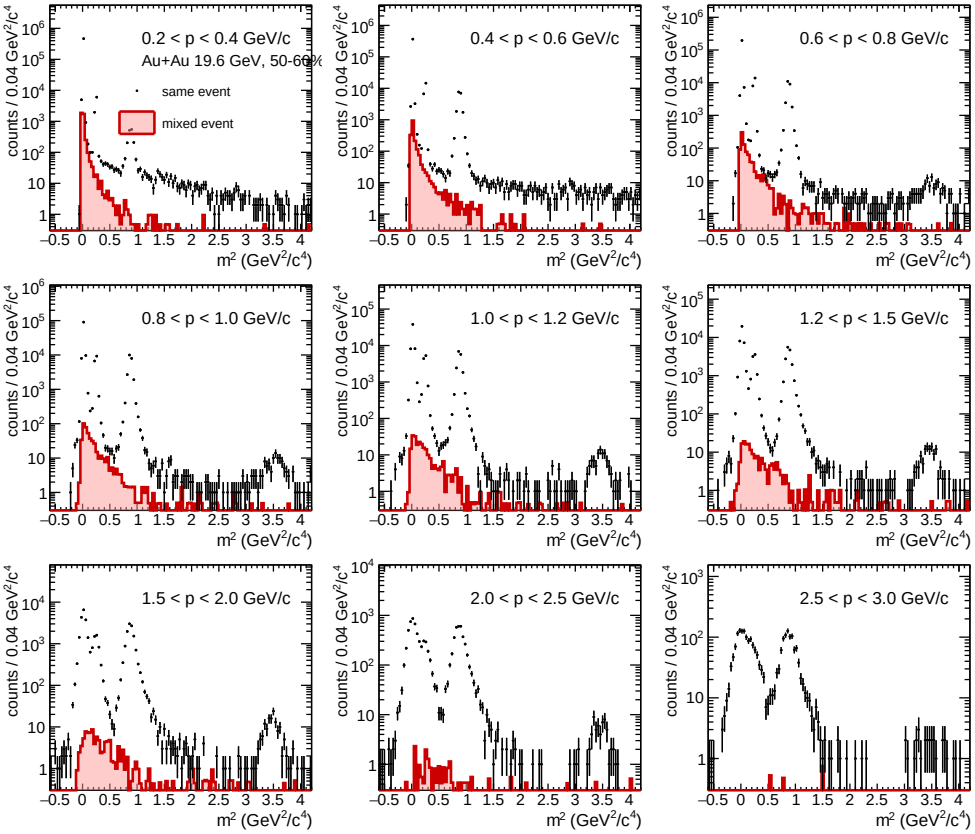
(a) m^2 of projected QA tracks, 20–30%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell$\pm$1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 29: SE vs ME m^2, 20–30%.



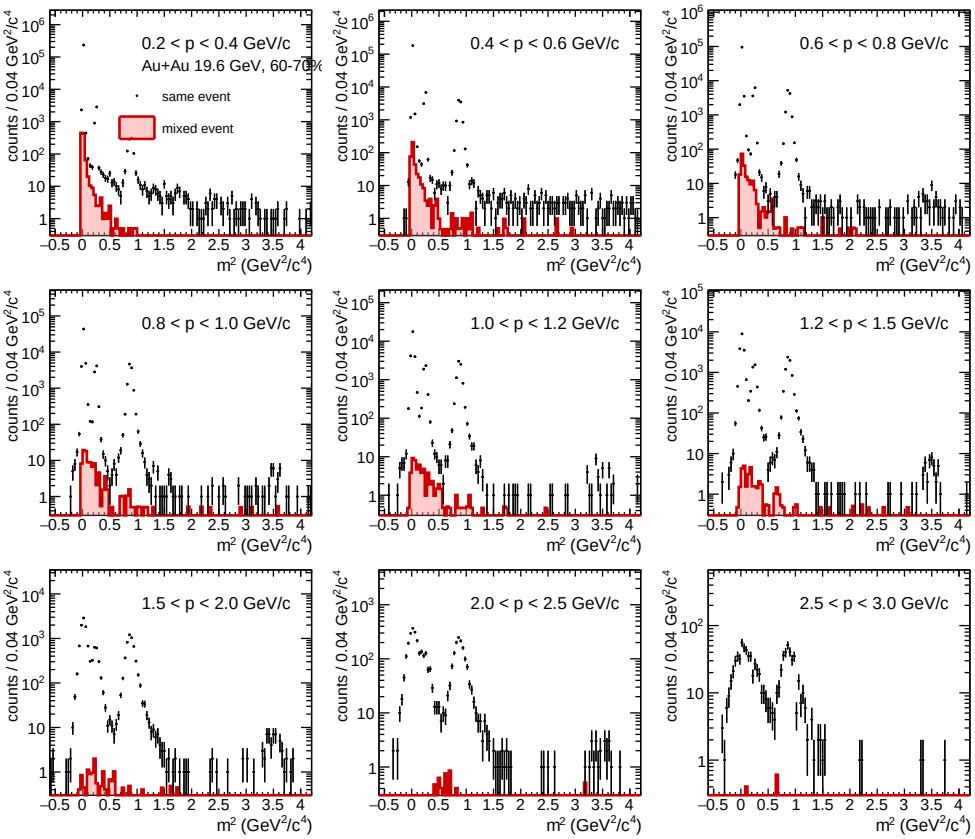
(a) m^2 of projected QA tracks, 40–50%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 30: SE vs ME m^2, 40–50%.



(a) m^2 of projected QA tracks, 50–60%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 31: SE vs ME m^2, 50–60%.



(a) m^2 of projected QA tracks, 60–70%, nine momentum bins (log scale, $0.04\,\mathrm{GeV}^2/c^4$ bins). Black points: same event (cell±1 rule, hit with time). Red filled: mixed event (overlay, all hits) normalized per projected track. Statistical errors on the SE points only.

Figure 32: SE vs ME m^2, 60–70%.